HYDRIS AI

HYDRIS FIELD GLASSES · DESIGN AND UX

Design System and UX Guidelines

The tokens, components, layouts and interaction rules that make every Hydris surface safe, glanceable and consistent in the field

DOCUMENT	VERSION	STATUS	DATE
HFG-DSN	1.0	Draft for review	October 2026

Document control

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Revision history

Version	Date	Author	Changes
1.0	October 2026	Design, Hydris	First edition: tokens and contrast audit per surface, component library, layout grids, interaction state machines, accessibility, content and governance

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Hydris Field Glasses is used by people standing next to running pumps, aeration basins and chemical dosing skids, often wearing gloves, ear defenders and safety glasses, sometimes in bright sun and sometimes in a dark pump room. The design system exists so that every surface (the marketing website, the operator console, the glasses heads-up app, the browser simulator and the native Android app) shows plant state the same way, uses colour for the same meaning, and never asks an operator to read more than they can take in at a glance.

This document describes the system **as it is built today**, with every token and component traced to the file that defines it, and it marks the gaps honestly. Where the code and the intended design differ, the difference is called out as a finding with a recommended fix. Status labels follow the house convention: **Built** (in the code and working), **Partial** (in the code with known gaps), **Planned** (designed or intended, not yet in the code).

5	3	4	19.05:1
surfaces website, console, glasses HUD, simulator, Android app	**token sources** globals.css, site.css brand layer, Android Theme.kt	**severity levels** critical, high, medium, low, one colour each	**primary text contrast** white on Deep Midnight (WCAG AAA)

1 Design principles

The principles below are ranked. When two of them conflict, the one higher in the list wins. Every design review and every QA pass (see the design QA checklist chapter) checks a screen against them in this order.

1.1 Safety first

Hydris is an information and records tool. It does not operate equipment, and the interface must never suggest that it does. That rule is visible in the product today:

- The glasses task screen states "Records only" and that Hydris does not operate equipment (`apps/web/src/glasses/GlassesApp.tsx`, `TaskDetail`). **Built.**
- The glasses Acknowledge action carries the inline hint "does not change equipment" (`AlertDetail`). **Built.**
- Recommended actions on alarms are labelled "Next step (approved procedure)" so the operator knows the text comes from an approved SOP, not from a model (`AlertDetail`). **Built.**
- Inspection runs show safety notices **before** the first step and render step cautions in the medium (amber) tone with a shield icon (`InspectionRunner`). **Built.**

> **IMPORTANT**
> No Hydris screen may present a control that looks like it starts, stops, opens, closes or sets plant equipment. Buttons are verbs about records: Acknowledge, Record, Submit, Resolve, Send for verification.

1.2 The one-glance rule

An operator should get the answer to "is anything wrong, and what do I do next?" from a single glance of under two seconds. In practice that means:

- One primary action per HUD screen and one focus indicator (`apps/web/src/glasses/hud.tsx` header comment). **Built.**

- Numbers are shown large, with their unit and their freshness, never as a bare value.
- Severity is carried by colour **and** by a word (Critical, High) and, on the HUD, by a border. Colour alone is never the only signal.
- Lists on the HUD are one column, short labels, with secondary meta in a smaller, dimmer line.

1.3 Advisory only

The voice assistant and the audit-scan OCR are advisory. Answers from "Hey Hydris" are grounded in the account's live data and shown with the facts (value, unit, label) they used (`apps/android/.../ui/screens/AskScreen.kt` shows up to four facts per answer). Scanned audit values always go to a human reviewer and then to a second approver in the console; the Android scan flow says so on its confirmation screen (`ScanScreen.kt`). The design must keep that distinction: model output is shown in Insight Purple contexts or with explicit "verify" copy, and is never styled as a confirmed record.

1.4 Never hide state

Every value has a state, and the state is always visible:

- **Freshness.** A reading is only meaningful together with when it was observed. The domain defines `live`, `delayed`, `stale` and `no_data` (`packages/domain/src/freshness.ts`), and missing data is reported as `no_data`, never as zero. The console renders this with `FreshnessBadge` (`apps/web/src/ui/index.tsx`). **Built.**
- **Connectivity.** The glasses app shows an offline strip ("Offline · showing data cached ...") and a queued-steps counter in the status bar (`GlassesApp.tsx`, `StatusBar`). **Built.**
- **Synthetic data.** Simulator-generated values carry a dashed purple "Synthetic" chip on every surface (`SyntheticBadge` in `ui/index.tsx`, the `synthetic` chip on Android alarm detail, "· synthetic" on the HUD), and the console shows a demo banner for demo tenants (`console/Shell.tsx`). **Built.**
- **Permissions.** When a UI profile blocks an action, the glasses app shows a dashed "not permitted by your profile" note instead of silently removing the control (`Blocked` in `GlassesApp.tsx`). **Built.**

1.5 Dark first for the glasses

Display glasses use an additive display: pure black renders as transparent. The HUD therefore uses a black canvas, light strokes, large type (at least 20 px for body content) and no large filled areas (`hud.tsx`). The glasses route is forced dark regardless of the user's theme (`FORCED_DARK_RE` in `apps/web/src/ui/theme-script.ts`). The console and website default to dark (`THEME_DEFAULT_MODE = 'dark'`) and offer a full light theme; the Android app is dark only today.

1.6 Consistency across surfaces

The same alarm should look like the same alarm on the console, the phone and the glasses: same severity colour family, same state words, same order of information (title, value and unit, asset, time, recommended action). The token chapter below shows where the surfaces currently drift and how to close the gaps.

2 Design system architecture

The system has four layers. Brand decisions flow down into tokens; tokens are implemented per platform; components consume only tokens; screens compose components.

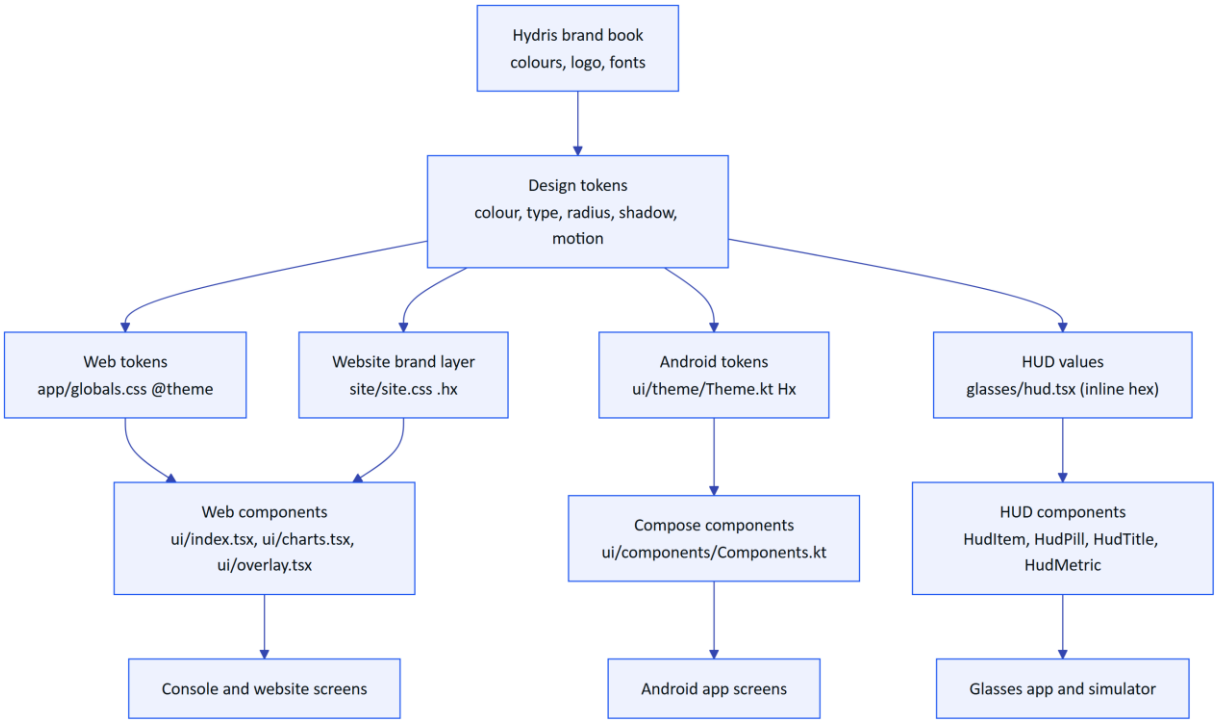


Figure 1. *Design system layers, from brand book to screens*

Table 1. Where the design system lives in the code

Layer	Web console and website	Glasses HUD and phone mode	Android app
Colour, radius, shadow, motion tokens	`apps/web/src/app/globals.css` (`@theme`, `:root`, `:root[data-theme='light']`)	Inherits globals.css; severity tints hard-coded in `apps/web/src/glasses/hud.tsx` and `GlassesApp.tsx`	`apps/android/app/src/main/java/ai/hydris/field/ui/ther` (`object Hx`)
Website brand overrides	`apps/web/src/site/site.css` (`.hx` scope)	Not used	Not used
Typography roles	`.t-display` to `.t-kpi-sm` in globals.css; Outfit via `next/font` in `app/layout.tsx`	Tailwind pixel sizes in `hud.tsx`	Material 3 `Typography` in Theme.kt
Components	`apps/web/src/ui/index.tsx`, `ui/charts.tsx`, `ui/overlay.tsx`, `ui/theme.tsx`	`hud.tsx` primitives plus screens in `GlassesApp.tsx`	`ui/components/Components.kt`
Icons	`lucide-react` 1.57.0	`lucide-react`	`material-icons-extended` (Outlined set)
Theme switching	`ui/theme.tsx`, `ui/theme-script.ts`	Forced dark	Dark only

Layer	Web console and website	Glasses HUD and phone mode	Android app
Shared token package	None	None	None

RISK

Tokens are defined three times and the HUD uses inline hex values. The values were aligned by hand, and several have drifted (see the severity mapping table). Until a single token source generates all three platforms (Planned, see the future chapter), every colour change must be made in all three files in the same pull request.

3 Brand and colour system

3.1 Brand palette

The brand palette comes from the Hydris AI brand book and is quoted in the header comments of `globals.css` and `site.css`. Core Blue is the dominant brand colour on the website; Signal Blue is the action colour in product surfaces. Deep Intelligence Blue (navy) is used for pressed states and deep brand panels.

Table 2. Brand palette

Name	Hex	Role	Where defined
Hydris Core Blue	`#1746E4`	Brand fills, hover on primary, chart series 1, website accent	`--color-brand` (globals.css), `--hx-blue` (site.css), `Hx.Core`
Deep Intelligence Blue	`#0C1E85`	Pressed primary, deep brand panels, logo navy	`--color-brand-deep`, `--hx-deep`
Signal Blue	`#1954F1`	Primary buttons, focus fills, selected state, links on paper	`--color-signal`, `--color-lime`, `Hx.Signal`
Signal tint (accent text)	`#6F93FF`	Accent **text** and focus ring on dark backgrounds	`--color-accent`, `Hx.SignalText`
Deep Midnight	`#081116`	App background (dark)	`--color-bg`, `Hx.Midnight`
Graphite Black	`#222222`	Print and brand graphite	`--color-graphite`
Pure White	`#FFFFFF`	Primary text on dark, ink on blue	`--color-text`
Alert Orange	`#FF2800`	Critical severity only	`--color-critical`, `--color-orange`, `Hx.Alert`
Active Green	`#00FE21`	OK and live: dots and text only, never large fills	`--color-ok`, `Hx.Ok`
Hydris Insight Purple	`#8A60F0`	AI, information, secondary data series, synthetic data	`--color-purple`, `--color-cobalt`, `Hx.Insight`

> **NOTE**
> The token names `lime`, `lime-2`, `lime-3` and `cobalt` are historical. In the code today `lime` means the brand blue role with white ink, and `cobalt` means Insight Purple (globals.css header comment). New code should use `signal`, `brand`, `brand-deep` and `purple`. Renaming the legacy roles is part of the token pipeline work.

3.2 Web neutral and surface tokens (dark and light)

The console defines every semantic token for both themes. The dark values live in `@theme`; the light values in `:root[data-theme='light']`. Components refer only to the semantic names, so a theme switch needs no JavaScript.

Table 3. Web surface and text tokens (apps/web/src/app/globals.css)

Token	Dark	Light	Semantic role
`--color-bg`	`#081116`	`#F6F8FC`	Page background
`--color-bg-2`	`#0C161C`	`#EEF2F8`	Inset areas, input wells

Token	Dark	Light	Semantic role
`--color-surface`	`#121A20`	`#FFFFFF`	Cards, menus, dialogs
`--color-surface-2`	`#1A2229`	`#F2F5FA`	Secondary buttons, chips, hover
`--color-surface-3`	`#252E35`	`#E5EAF2`	Pressed, disabled primary, progress track
`--color-line`	white 7.5 %	slate 8.5 %	Hairline borders
`--color-line-2`	white 14 %	slate 15 %	Strong borders, inputs
`--color-text`	`#FFFFFF`	`#1B2129`	Primary text
`--color-text-2`	`#C5CCD2`	`#46505B`	Secondary text
`--color-muted`	`#8D979F`	`#66717D`	Labels, meta, captions
`--color-paper`	`#EEF1F4`	`#FFFFFF`	KPI paper cards (light island on dark page)
`--color-ink` / `--color-ink-2`	`#081116` / `#4C5862`	`#1B2129` / `#545F6B`	Text on paper
`--color-accent`	`#6F93FF`	`#1746E4`	Accent text, focus ring
`--color-scrim`	midnight 72 %	slate 38 %	Modal backdrop
`--color-field`	`#0C161C`	`#FFFFFF`	Input background
`--color-placeholder`	`#67727B`	`#8A949F`	Input placeholder
`--color-chart-grid`	white 7 %	slate 7 %	Chart gridlines

3.3 Android tokens

The Android app maps the same palette into a Material 3 dark colour scheme (`Theme.kt`). `primary` is Signal Blue, `secondary` is Insight Purple, `tertiary` is a softer green (`OkSoft`), `error` is Alert Orange.

Table 4. Android Hx tokens (apps/android/.../ui/theme/Theme.kt)

Token	Hex	Material role	Use
`Hx.Midnight`	`#081116`	`background`, `surfaceContainerLow`	Screen background
`Hx.Surface`	`#0E1A21`	`surface`, `surfaceContainer`	Cards, navigation bar
`Hx.Surface2`	`#13222B`	`surfaceVariant`, `surfaceContainerHigh`	Raised containers
`Hx.Line`	white 12 %	`outline`	Card borders, progress track
`Hx.Signal`	`#1954F1`	`primary`	Primary buttons, progress, spinner
`Hx.Core`	`#1746E4`	`primaryContainer`	Pressed and selected
`Hx.SignalText`	`#6F93FF`	(direct)	Accent text, selected tab
`Hx.Alert`	`#FF2800`	`error`	Critical severity
`Hx.High`	`#F26414`	(direct)	High severity, stale data, error notes
`Hx.Medium`	`#F5B83D`	(direct)	Medium severity, delayed data
`Hx.Ok`	`#00FE21`	(direct)	Active Green (reserved)
`Hx.OkSoft`	`#34D27A`	`tertiary`	Live data, typical-range band
`Hx.Insight`	`#8A60F0`	`secondary`	AI, synthetic chip

Token	Hex	Material role	Use
`Hx.Text` / `Hx.Text2` / `Hx.Muted`	`#F2F5F7` / `#B7C2CA` / `#7D8B95`	`onBackground`, `onSurfaceVariant`	Text hierarchy

3.4 HUD colour values

The HUD inherits `globals.css`, so `lime` (focus border, selected pill) is Signal Blue `#1954F1`. Severity tints on the HUD are lighter than the console's, because they must read as thin strokes and text on an additive display. They are currently inline hex values in `hud.tsx` and `GlassesApp.tsx`.

Table 5. HUD colour values (apps/web/src/glasses/hud.tsx, GlassesApp.tsx)

Use	Border	Text	Notes
Critical	`#FF5A4E`	`#FF8A80`	`HudPill tone="critical"`, `AlertInterrupt` border
High	`#F26414`	`#FF9A5C`	Matches Android `Hx.High` border
Medium	`#F5B83D`	`#F7C96A`	Step cautions, safety notices icon
OK	`#4ED98E`	`#7FE6AE`	Softer than Active Green to avoid glare
Neutral line	white 25 %	white 80 %	Default pill
Focus / selected	`lime` (`#1954F1`)	`#141714` on fill	See the contrast findings below
Critical banner background	n/a	n/a	`#2A0D0B` (critical), `#2A1608` (high and offline strip)

3.5 Severity and status colour mapping

Severity colours are semantic and are **never** reused as decoration (globals.css header). The table shows how each surface renders each severity today, which is where the drift is easiest to see.

Table 6. Severity colour mapping by surface

Severity	Web console fill / text (dark)	Web light text	Website `.hx`	Android `Hx.severity()`	HUD border / text
Critical	`#FF2800` / `#FF7A5C`	`#C41F00`	`#FF2800`	`#FF2800`	`#FF5A4E` / `#FF8A80`
High	`#FF9F1C` / `#FFB547`	`#A35200`	`#FF5A1F`	`#F26414`	`#F26414` / `#FF9A5C`
Medium	`#F5C542` / `#F7D36E`	`#7F6000`	(inherits)	`#F5B83D`	`#F5B83D` / `#F7C96A`
Low	`#8FB0FF` / `#B3C9FF`	`#1746E4`	(inherits)	`#6F93FF`	neutral line
OK / live	`#00FE21` / `#00FE21`	`#0B7A1F`	`#00FE21`	`#34D27A` (live), `#00FE21`	`#4ED98E` / `#7FE6AE`

> **DECISION**
>
> The canonical High colour is **`#F26414`** (the brand-adjacent orange already used on Android and the HUD). The console's amber `#FF9F1C` and the website's `#FF5A1F` will move to it, with a lighter text tint for small text on dark. Critical stays Alert Orange `#FF2800` everywhere, with surface-specific text tints for legibility. Status: **Planned** (token alignment).

The same colours drive object states through `StateChip` (`ui/index.tsx`). The mapping rule: red family for "needs attention now", amber for "seen, not finished", purple for "someone is working on it", green for "done or healthy", outline for "neutral or historical".

Table 7. State to tone mapping used by StateChip (apps/web/src/ui/index.tsx)

Tone	States
Critical	`active` (alarm), `down`, `revoked`, `urgent`
Medium	`acknowledged`, `paused`, `on_hold`, `degraded`, `disconnected`
Cobalt (Insight Purple)	`investigating`, `in_progress`, `assigned`, `maintenance`
OK	`resolved`, `approved`, `completed`, `operational`, `connected`
High	`rejected`, `suspended`
Brand blue (lime)	`submitted`
Outline	`closed`, `scheduled`, `cancelled`, `open`, `standby`, `draft`, `archived` and other neutral states

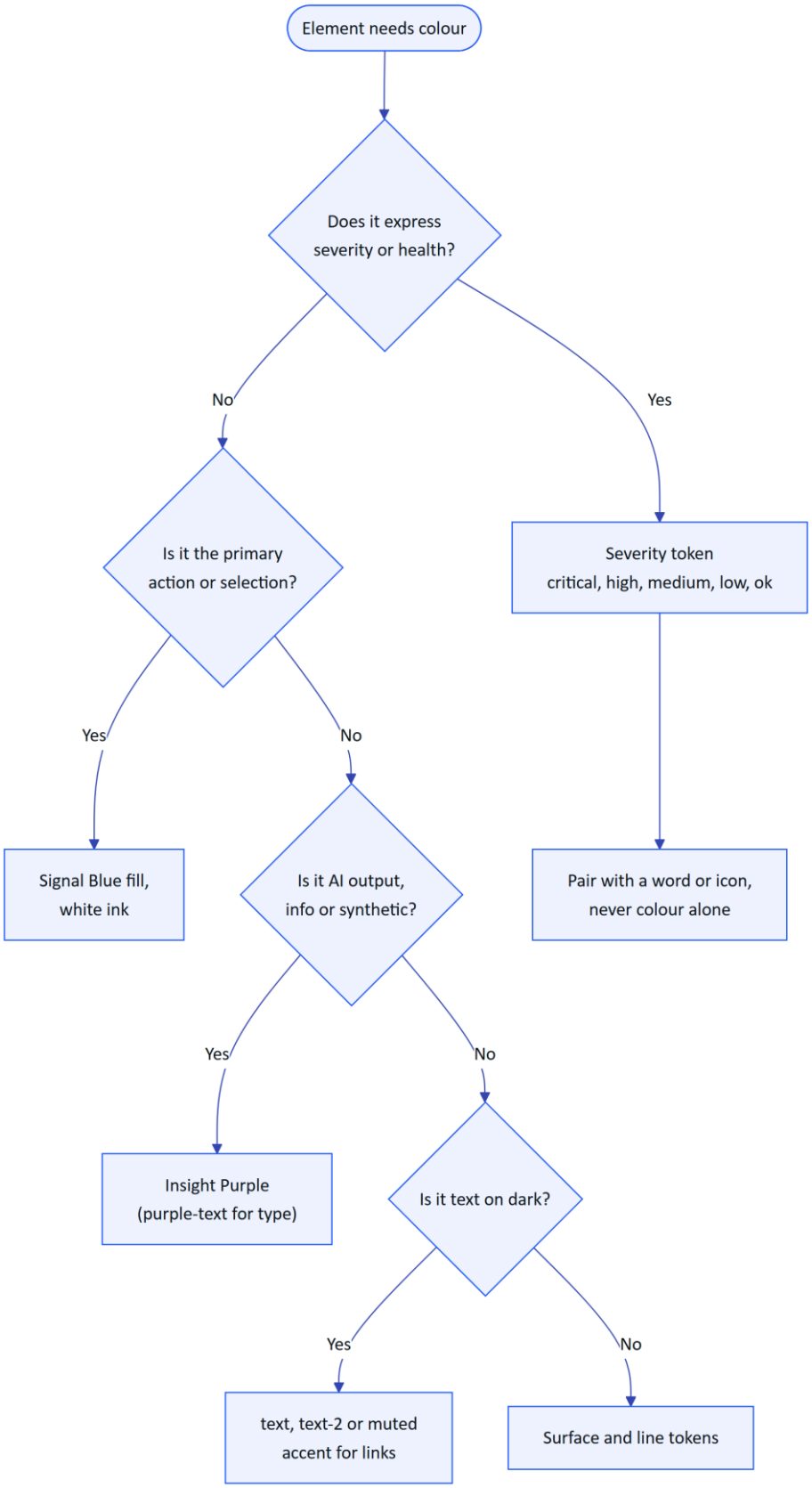


Figure 2. *Choosing a colour for a value or element*

3.6 Contrast audit

Contrast ratios below were computed with the WCAG 2.x relative luminance formula from the hex values in the code (script run during authoring; alpha tints were not composited and are excluded). The target for all product text is WCAG 2.2 **AA**: 4.5:1 for normal text and 3:1 for large text (24 px regular or 18.66 px bold and above) and for non-text UI components.

Table 8. Contrast ratios for key colour pairs (computed from code values)

Foreground	Background	Ratio	Result	Where used
White `#FFFFFF`	Midnight `#081116`	19.05:1	AAA	Primary text, console and HUD
text-2 `#C5CCD2`	surface `#121A20`	10.84:1	AAA	Secondary text in cards
muted `#8D979F`	surface `#121A20`	5.91:1	AA	Labels, meta
muted `#8D979F`	bg `#081116`	6.41:1	AA	Captions on page
Signal `#1954F1`	Midnight `#081116`	3.25:1	Large text only	Why accent text uses `#6F93FF`
Accent `#6F93FF`	Midnight `#081116`	6.62:1	AA	Links, selected tab, focus ring
White	Signal `#1954F1`	5.87:1	AA	Primary button label
White	Core Blue `#1746E4`	6.92:1	AA	Hover primary, brand panel
White	Deep Blue `#0C1E85`	13.64:1	AAA	Pressed primary
White	Danger solid `#E02400`	4.75:1	AA	`btn-danger-solid`
critical-text `#FF7A5C`	surface `#121A20`	6.86:1	AA	Critical chip text
high-text `#FFB547`	surface	10.01:1	AAA	High chip text
ok-text `#00FE21`	surface	12.73:1	AAA	OK chip text
purple-text `#B69CFF`	surface	7.69:1	AAA	Insight and synthetic chips
Ink `#081116`	paper `#EEF1F4`	16.81:1	AAA	KPI paper cards
Light text `#1B2129`	White	16.20:1	AAA	Light theme body
Light muted `#66717D`	bg `#F6F8FC`	4.68:1	AA	Light theme labels
Light critical `#C41F00`	White	5.93:1	AA	Light theme critical text
Light high `#A35200`	White	5.58:1	AA	Light theme high text
Light ok `#0B7A1F`	White	5.50:1	AA	Light theme OK text
Android Text2 `#B7C2CA`	Surface `#0E1A21`	9.75:1	AAA	Secondary text
Android Muted `#7D8B95`	Surface `#0E1A21`	5.05:1	AA	Meta lines
Android Insight `#8A60F0`	Surface `#0E1A21`	4.22:1	Fails AA (normal)	Synthetic chip text
White	Android Alert `#FF2800`	3.78:1	Fails AA (normal)	Filled critical chip, 12 sp
White	Android High `#F26414`	3.18:1	Fails AA (normal)	Filled high chip, 12 sp
HUD critical `#FF8A80`	Black	9.20:1	AAA	HUD critical text
HUD white 55 % (about `#8C8C8C`)	Black	6.25:1	AA	HUD meta lines
HUD ink `#141714`	Signal `#1954F1`	3.08:1	Large text only	Selected pill, View button, digit entry

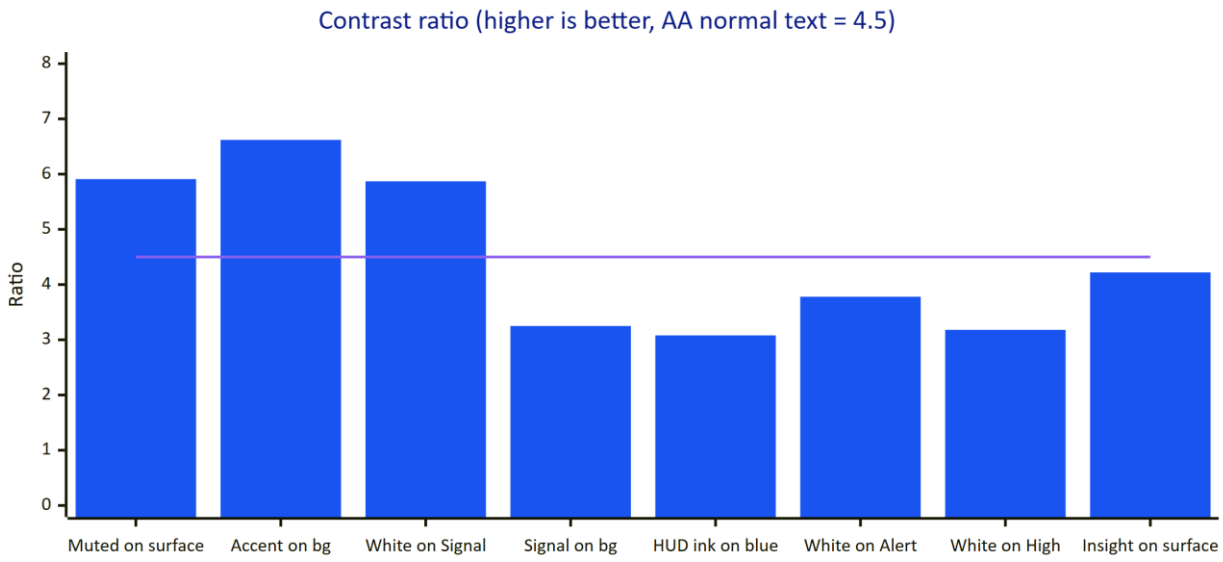


Figure 3. *Contrast of selected text pairs against the 4.5:1 AA threshold*

WARNING

Three contrast failures exist in shipped code. (1) **HUD selected pill and View button**: `#141714` ink on Signal Blue is 3.08:1. The ink colour is a leftover from when the `lime` role was a light lime green. Fix: white ink on the blue fill (5.87:1) in `HudPill tone="lime"`, `AlertInterrupt` and the digit-entry cell in `GlassesApp.tsx`. (2) **Android filled severity chips**: white on `#FF2800` (3.78:1) and on `#F26414` (3.18:1) at 12 sp. Fix: use the tinted chip style (coloured text on a 14 % fill) or dark ink on the fill. (3) **Android synthetic chip**: Insight `#8A60F0` text on Surface is 4.22:1. Fix: add an `Hx.InsightText` tint equivalent to the web `#B69CFF`. Status: **Planned** fixes, tracked in the QA checklist.

4 Typography

4.1 Typefaces

Table 9. Typefaces by surface

Surface	Typeface in code	Source	Status
Console, website, glasses web app, simulator	Outfit (300, 400, 500, 600)	`next/font/google` in `apps/web/src/app/layout.tsx`, exposed as `--font-outfit`	**Built**
Monospace (IDs, codes)	System UI monospace stack	`--font-mono` in globals.css	**Built**
Android app	Platform default (Roboto on most devices)	Material 3 `Typography` without a custom `FontFamily`	**Built**
Word and PDF product documents	Arial and Calibri	`docs/product/tools/build.js` theme	**Built**
Brand typefaces Unison Pro (display) and Eloquia Text (body)	Not in any product surface	Brand book	**Planned** (licensing for web and app embedding)

> **DECISION**
>
> Product UI keeps Outfit and the platform font until web and app embedding licences for Unison Pro and Eloquia Text are confirmed. When they are, Unison Pro is limited to display roles (`t-display`, `t-hero`, website headings) and Eloquia Text replaces Outfit for body roles. Numerals in KPIs keep tabular figures (`.num`), whichever face is used.

All numeric values use tabular, lining figures (`.num` in globals.css: `font-variant-numeric: tabular-nums lining-nums`) so live values do not jitter as digits change. HUD metrics and digit-entry cells apply `.num` too.

4.2 Web type scale

Table 10. Web typography roles (apps/web/src/app/globals.css)

Role class	Size	Line height	Tracking	Weight	Use
`t-display`	clamp 44 to 92 px	1.02	-0.04 em	600	Website hero
`t-hero`	clamp 36 to 64 px	1.04	-0.035 em	600	Website section heroes
`t-page`	clamp 30 to 44 px	1.06	-0.03 em	600	Console page titles
`t-section`	21 px	1.25	-0.02 em	600	Card groups, dialog titles
`t-card`	16.5 px	1.3	-0.012 em	600	Card titles
body (`.input`, table)	14 to 14.5 px	1.5	normal	400	Body text, tables, inputs
`t-label`	13 px	1.35	0.005 em	400	Field labels, KPI labels
`t-eyebrow`	11.5 px	1.4	0.16 em, uppercase	500	Section eyebrows
`t-kpi`	clamp 38 to 54 px	0.95	-0.045 em	300	KPI numerals
`t-kpi-sm`	28 px	1.05	-0.035 em	300	Secondary KPI numerals

Role class	Size	Line height	Tracking	Weight	Use
Table header	11.5 px	n/a	0.08 em, uppercase	500	`.table th`

4.3 Android type scale

Table 11. Android typography (apps/android/.../ui/theme/Theme.kt)

Material role	Size	Weight	Tracking	Use
`displaySmall`	34 sp / 38 sp	SemiBold	-0.8 sp	Large greetings
`headlineSmall`	24 sp	SemiBold	-0.4 sp	Screen titles ("Audit scan", "Glasses")
`titleLarge`	20 sp	SemiBold	-0.2 sp	Section titles
`titleMedium`	16 sp	Medium	0	List item titles, option rows
`bodyLarge`	16 sp / 22 sp	Regular	0	Primary body
`bodyMedium`	14 sp / 20 sp	Regular	0	Secondary body, KPI labels
`bodySmall`	12.5 sp / 17 sp	Regular	0	Meta lines
`labelSmall`	11 sp	Medium	1.4 sp	Eyebrows (uppercase)
KPI value (inline)	34 sp	SemiBold	-1 sp	`KpiTile` value
Alarm reading (inline)	36 sp	SemiBold	0	Alarm detail reading

Sizes are in `sp`, so the app follows the operator's system font-size setting. **Built.**

4.4 HUD type scale

The HUD is designed on a 600 by 600 canvas. All sizes below are canvas units, scaled to the display.

Table 12. HUD typography (apps/web/src/glasses/hud.tsx, GlassesApp.tsx)

Role	Size	Weight	Use
`HudTitle`	38	Regular, -0.03 em	Screen title
`HudMetric` value	52	Light, tabular	Key reading
Digit entry cell	56	Light, tabular	Inspection numeric input
Alert interrupt title	34	Regular	Full-screen alarm
`HudItem` row	23	Regular	List rows, actions
Body and instructions	19 to 22	Regular	Step instructions, SOP text
Status bar	18	Regular	Connection, queue, operator
`HudPill`	18	Medium	Status pills
Eyebrow	17	Uppercase, 0.14 em	Section and step labels
Meta line	17	Regular, white 55 %	Secondary row text
Hint bar	16	Regular, white 45 %	Control hints

> **TIP**
>
> The HUD comment sets the floor at 20 px for content. Meta lines, eyebrows and hints at 16 to 18 are allowed because they are secondary and never carry the only copy of a safety-relevant value. A UI profile can raise everything with `layout.textScale` (0.9, 1, 1.15, 1.3; clamped to 0.85 to 1.4) in `packages/domain/src/uiprofile.ts`; the glasses app applies it with CSS `zoom` in `GlassesApp.tsx`. **Built.**

5 Spacing, radius and elevation

5.1 Spacing

Web layouts use the Tailwind 4 spacing scale (4 px base) without custom spacing tokens. The recurring values are documented here so new screens stay on the rhythm.

Table 13. Spacing rhythm in use

Context	Web console	Android	HUD
Page gutter	16 px mobile, 32 px from `md` (`px-4 md:px-8`)	16 dp (`padding(horizontal = 16.dp)`)	32 units HUD, 24 units phone mode (`px-8` / `px-6`)
Card padding	24 px (`.card`), 18 px tight, 20 to 24 px KPI	16 dp default, 14 dp list cards	16 to 24 units
Gap between cards	12 to 16 px	12 dp (`Row2`, `spacedBy(12.dp)`)	12 units
Section top spacing	32 px (`pt-8`)	20 dp (`SectionTitle`)	20 units (`HudTitle` margin)
Max content width	1440 px (`max-w-[1440px]`)	Full width	600 units (HUD) / 540 units (phone)

5.2 Radius

Table 14. Radius tokens

Token	Value	Use
`--radius-xs`	6 px	Small tags, kbd
`--radius-sm`	10 px	Skeletons
`--radius-md`	14 px	Inputs, selects
`--radius-lg`	18 px	Inset cards, menus
`--radius-xl`	24 px	Cards, panels, empty states
`--radius-2xl`	32 px	Centered dialogs
Pill	9999 px	Buttons, chips, capsules, icon buttons
Android `CardShape`	22 dp	`HCard`
Android chip / progress	50 %	`Chip`, `ProgressBar`
HUD row	26 units	`HudItem`
HUD alert	34 units	`AlertInterrupt`

5.3 Elevation

Elevation on dark surfaces comes from a top-edge sheen and graded surface colours more than from shadows.

Table 15. Elevation tokens (globals.css)

Token	Dark	Light	Use
`--sheen`	inset 1 px white 4.5 %	soft 1 to 2 px shadow	Every card and capsule
`--shadow-1`	1 px plus 8 px blur, black 20 to 30 %	slate 5 %	Resting raised elements
`--shadow-2`	6 px plus 28 px, black 30 to 35 %	slate 6 to 9 %	Hover lift, glass panels
`--shadow-3`	12 px plus 72 px, black 35 to 55 %	slate 8 to 16 %	Menus, dialogs, mobile tab bar
Brand panel glow	18 px 48 px Core Blue 22 %	18 %	`.card-lime` focal panel

Android cards are flat (`tonalElevation = 0` on the navigation bar, no shadow on `HCard`), separated by a 1 dp `Hx.Line` border. The HUD uses no shadows at all, since shadows are invisible on an additive display.

6 Iconography

Table 16. Icon systems

Surface	Library	Style	Default sizes	Status
Console and website	`lucide-react` 1.57.0	1.5 to 2 px stroke, rounded	19 px navigation, 16 to 18 px buttons, 14 px inline	**Built**
Glasses HUD and phone mode	`lucide-react`	Same	22 to 28 units	**Built**
Android	`androidx.compose.material:material-icons-extended`	Outlined set (`Icons.Outlined.*`, `Icons.AutoMirrored.Outlined.*`)	18 to 24 dp	**Built**

The two libraries are visually close (outlined, geometric), which keeps the surfaces consistent without sharing assets. Use the same concept for the same meaning:

Table 17. Icon concept mapping across surfaces

Concept	Lucide (web, HUD)	Material Outlined (Android)
Home / overview	`LayoutGrid`, `Glasses` (HUD home tab)	`Dashboard`
Alarms	`Siren` (console nav), `AlertTriangle` (HUD)	`Warning`
Inspections	`ClipboardCheck`	`FactCheck`
Maintenance	`Wrench`	`Build`
Audit scan	`ScanLine`	`DocumentScanner`
Voice / Hydris AI	`Mic`, `Sparkles`	`AutoAwesome`, `Mic`, `MicOff`
Glasses / devices	`Glasses`	`Visibility`
Back	`ChevronLeft`	`AutoMirrored.Outlined.ArrowBack`
Camera	n/a	`PhotoCamera`
Safety notice	`ShieldAlert`	n/a
Not permitted	`Ban`	n/a

Rules: icons never stand alone for an action on the HUD or phone (always with a label); icon-only buttons on the web carry an `aria-label` and a `Tooltip` (`IconButton` in `ui/index.tsx`); decorative icons are `aria-hidden`.

7 Component library

7.1 Component inventory

Table 18. Component inventory

Component	Purpose	Surfaces	States	Status
`Button`, `ButtonLink`	Actions: primary, secondary, ghost, light, cobalt, danger, danger-solid	Web	default, hover, active, focus, disabled, busy (`aria-busy`)	**Built**
`IconButton`	Circular icon actions with label and badge	Web	default, light, lime, ghost, outline, on-paper, danger; sm, md, lg	**Built**
Material `Button`, `OutlinedButton`	Primary and secondary actions, 52 to 54 dp tall	Android	enabled, disabled (busy)	**Built**
`Card`	Section container (dark, paper, lime focal)	Web	default, tight, hover lift	**Built**
`HCard`	Card with optional click and accent border	Android	default, clickable, accent	**Built**
`KpiCard`	KPI island: label, value plus unit, sub line, open link	Web	paper, dark, lime; critical, high, ok accent	**Built**
`KpiTile`	KPI tile: label, 34 sp value, sub line, icon	Android	accent colour, clickable	**Built**
`HudMetric`	Large reading with unit and label	HUD	tone colour	**Built**
`Chip`, `SeverityChip`, `StateChip`	Status and category labels	Web	10 tones including synthetic	**Built**
`Chip` (Compose)	Status label	Android	tinted, filled	**Partial** (filled contrast)
`HudPill`	Status pill on HUD	HUD	line, lime, critical, high, ok, medium	**Partial** (lime contrast)
`FreshnessBadge`	Live, delayed, stale, no data with time	Web	4 freshness states, pulsing live dot	**Built**
`SyntheticBadge`	Marks simulator data	Web, HUD text, Android chip	n/a	**Built**
`Capsule`	Segmented control and filters	Web	selected, hover, counts, lime variant	**Built**
`Input`, `PasswordInput`, `Select`, `Textarea`, `Field`	Forms with label, hint, error	Web	hover, focus, invalid, disabled	**Built**
`Switch`	Boolean settings	Web	on, off, disabled	**Built**
`DataTable`	Sortable, clickable tables	Web	hover, focus, selected, dense, empty	**Built**
`Timeline`	Event history	Web	lime, critical, ok, muted, cobalt	**Built**
`KeyValues`	Definition grid	Web	1 to 3 columns	**Built**

Component	Purpose	Surfaces	States	Status
`Progress`	Progress bar with ARIA	Web	lime, ok, critical	**Built**
`ProgressBar`	6 dp progress bar	Android	colour	**Built**
`Dialog`, `ConfirmDialog`	Modal, side panel, bottom sheet; confirmation with optional required note	Web	open, destructive, note required	**Built**
Toasts (`ToastProvider`, `useToast`)	Transient confirmations with timer	Web	info, error	**Built**
`EmptyState`, `ErrorState`, `LoadingBlock`, `Skeleton`, `QueryView`	Data states	Web	empty, not found, forbidden, error with retry, loading	**Built**
`Loading`, `ErrorNote`	Data states	Android	spinner, error note	**Built**
`HudItem`	Focusable HUD row	HUD	active (focus), disabled, tone	**Built**
`HudTitle`	Screen title with eyebrow	HUD	n/a	**Built**
`PhoneTabBar`	Bottom tab bar in phone mode	Glasses phone mode	active, back, badge	**Built**
`HintBar`	Control hints aligned to handedness	HUD	left, right	**Built**
`AlertInterrupt`	Full-screen critical or high alarm	HUD	critical, high, synthetic	**Partial** (button contrast)
`GuidedCapture`	Live A4 framing with guidance and auto-capture	Glasses, web	8 guidance codes, ready progress	**Built**
Web charts (`ui/charts.tsx`)	Sparkline, Sparkbars, BarStrip, Gauge, RadialGauge, Donut, StackedBar, BarList, Heatmap, BulletBar, RangeBar, Delta, MiniTrend, Funnel, StatusStrip, TrendChart	Web	empty state, screen-reader table	**Built**
Android charts	`Sparkline`, `LineChart`, `Bars`, `Donut` (Canvas)	Android	fewer than 2 points draws nothing	**Partial** (no accessible description)
`ThemeToggle`	System, light, dark	Web	checked, compact	**Built**
Haptic feedback	Confirm critical actions by vibration	Android, glasses	n/a	**Planned**

7.2 Cards

Web `Card` (`ui/index.tsx`, `.card` in globals.css): surface background, 1 px line border, 24 px radius, 24 px padding, sheen. Three tones: default dark; `paper` (light island on a dark page, used for KPIs so numbers pop); `lime`, the focal brand panel, a radial gradient of Core Blue with white type. Use at most **one** focal panel per view. Interactive cards opt into `.lift` (2 px rise on hover, settle on press).

Android `HCard` (`Components.kt`): `Hx.Surface`, 22 dp radius, 1 dp `Hx.Line` border, 16 dp padding. Passing `accent` tints the border at 55 % alpha; the home screen uses this for critical alarm cards and the alarm detail uses a Signal accent for the recommended action card.



Figure 4. *Android home: KPI tiles, accent-bordered alarm cards and sparkline rows*

7.3 KPI tiles

Table 19. KPI tile specification

Property	Web `kpiCard`	Android `kpiTile`	HUD `hudMetric`
Minimum height	124 px, 148 px from `sm`	Content height	Content height
Label	13.5 px, 75 % opacity, top left	`bodyMedium`, Text2, single line	17, white 60 %, below value
Value	`t-kpi` (38 to 54 px, light 300)	34 sp SemiBold	52 light
Unit	15 px, 60 % opacity, baseline aligned	Inline in value or sub	20, white 60 %

Property	Web `KpiCard`	Android `KpiTile`	HUD `HudMetric`
Accent	`kpi-accent-critical`, `-high`, `-ok`; deepened on paper for AA	`accent` colours the value	`tone` colours the value
Link	`OpenLink` arrow, top right	Whole tile clickable	Not interactive
Sub line	13 px, 70 % opacity	`bodySmall`, Muted	n/a

Rules: a KPI value is a fact with a unit and a time context ("in the last 24 h"); never show a KPI computed from stale data without saying so; never use the accent colour unless the number itself is the alert.

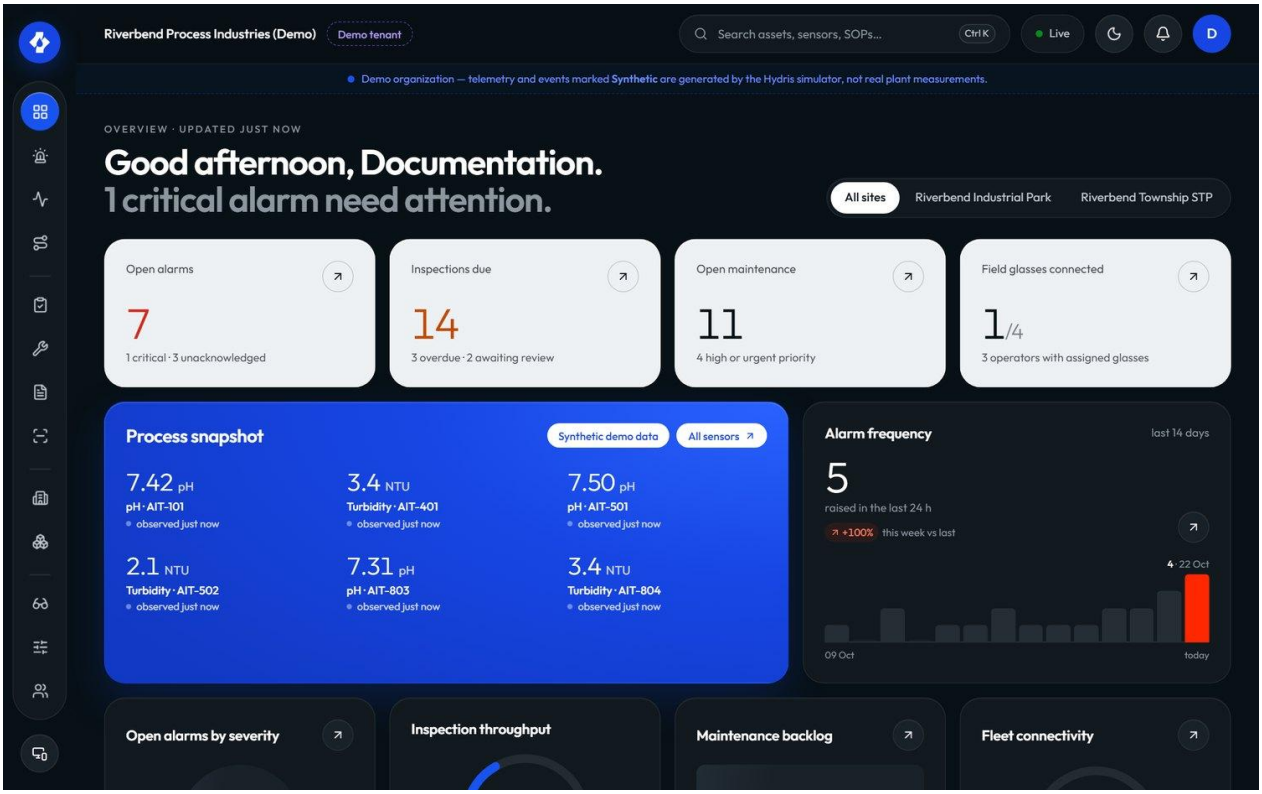


Figure 5. *Console overview: paper KPI islands, focal blue panel and live charts*

7.4 Chips and badges

Web chips are 28 px tall pills, 12.5 px medium text, with a 1 px tinted border and a 12 to 15 % tinted fill per tone. Clickable chips get a 44 px hit area through an `::after` pseudo-element. `SeverityChip` adds a dot in the severity colour plus the word. `chip-synthetic` is a dashed purple outline so simulated data is recognisable even in greyscale.

Android `Chip` is 12 sp medium text with 10 by 4 dp padding on a 14 % tint, or a solid fill when `filled = true` (alarm severity on the alarm detail screen). The HUD `HudPill` is 34 units tall with a 2 unit border and 18 unit text; on the HUD, outlines are preferred to fills.

7.5 Buttons

Table 20. Button specification (globals.css)

Variant	Fill	Text	Hover	Use
`btn-primary`	Signal `#1954F1`	White	Core Blue plus blue glow; pressed Deep Blue	One per view: the main record action

Variant	Fill	Text	Hover	Use
`btn-secondary`	surface-2	text	surface-3	Secondary actions
`btn-ghost`	transparent	text-2	surface-2	Tertiary, toolbars
`btn-light`	White	ink	paper	On the blue focal panel
`btn-cobalt`	Core Blue	White	Deep Blue	Brand emphasis on the website
`btn-danger`	transparent, orange border	critical-text	12 % orange fill	Reversible destructive actions
`btn-danger-solid`	`#E02400`	White	`#C41F00`	Irreversible destructive confirmation

Sizes: `btn-sm` 36 px (with an invisible 44 px hit area), default 44 px, `btn-lg` 52 px. All buttons scale to 0.97 on press, show 50 % opacity when disabled (the disabled primary turns into a neutral surface so it reads as "not yet" rather than a faded blue), and set `aria-busy` while an action is in flight. Android primary actions are Material `Button` at 52 to 54 dp height and full width in task flows (alarm detail, scan review), with `OutlinedButton` for secondary actions.

7.6 Inputs and forms

Inputs are at least 46 px tall, 14 px radius, 14.5 px text, on `--color-field` with a `line-2` border. Focus shows a Signal Blue border plus a 4 px Signal glow at 22 %; invalid inputs show a critical border and a `field-error` line prefixed by a small critical dot (so the error is not colour-only). `Field` always renders a visible label, an optional hint and an error, linked by `id`. Glove-friendly rules: never rely on small inline controls in the field apps; on Android use full-width text fields (the Ask screen uses a 26 dp radius pill field) and full-width buttons.

7.7 Tables

`.table` (globals.css) and `DataTable` (`ui/index.tsx`): sticky uppercase 11.5 px headers, 13 px row padding (10 px dense), hover wash, a 2 px Signal inset on the first cell of clickable rows, keyboard focus outline, and a 14 % Signal fill for the selected row. Sorting uses `aria-sort` on the header. Tables belong to the console only; phones and the HUD use lists.

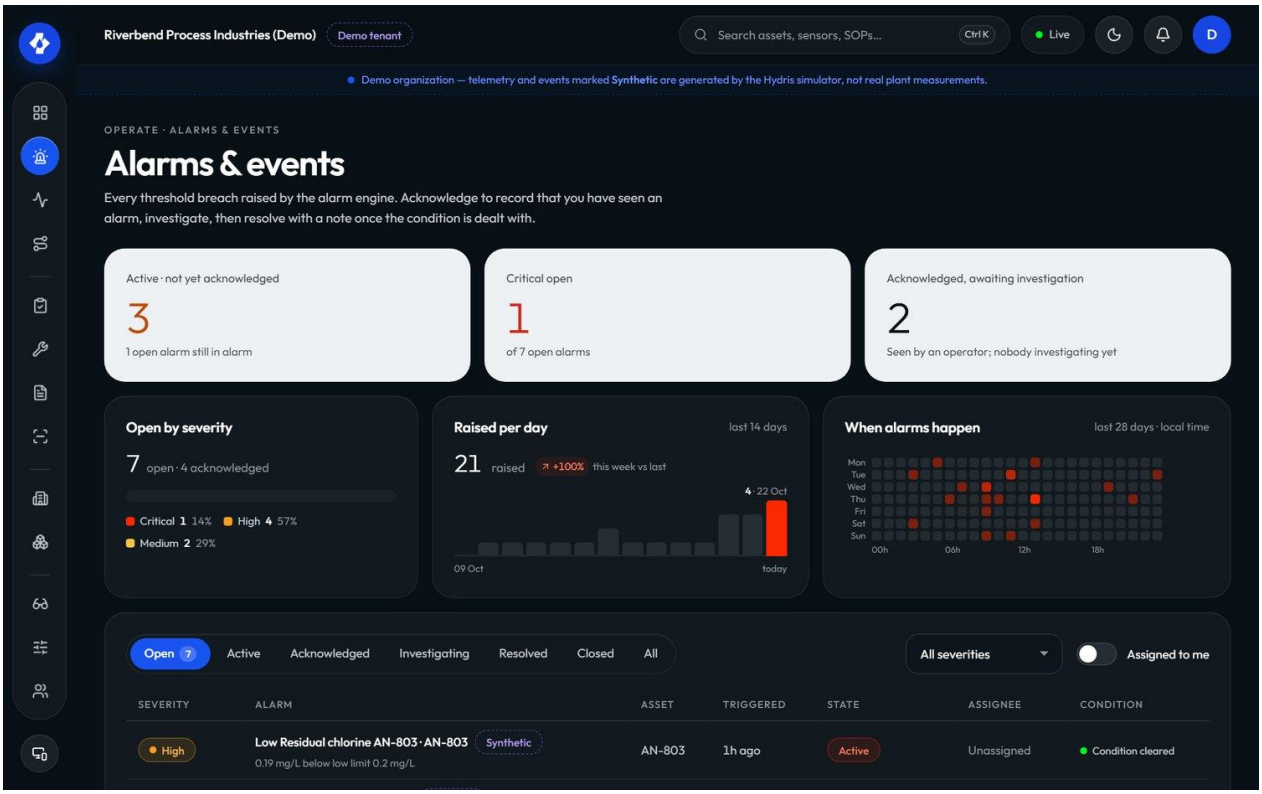


Figure 6. *Console alarms list: severity chips, state chips and clickable rows*

7.8 Alarm rows

An alarm row shows, in this order: severity (chip or coloured border plus word), title, value with unit, asset tag, state, and time since it was raised. Synthetic alarms add the synthetic marker.

Table 21. Alarm row anatomy by surface

Element	Console table row	Android list card	HUD `HudItem`
Severity	`SeverityChip` (dot plus word)	Accent border (critical) plus severity colour	`tone` colours the text, pill on detail
Title	Primary text	`titleMedium`	23 unit row text
Value and unit	Column	Meta line	Detail screen, 24 unit
State	`StateChip`	Chip on detail	Meta line ("active · 3m ago")
Time	Relative	`ago()` (just now, 5m ago)	`timeAgo`
Synthetic	`SyntheticBadge`	Purple chip	"· synthetic"
Open	Row click	Card click to `alarm/{id}`	Select to `alert` screen

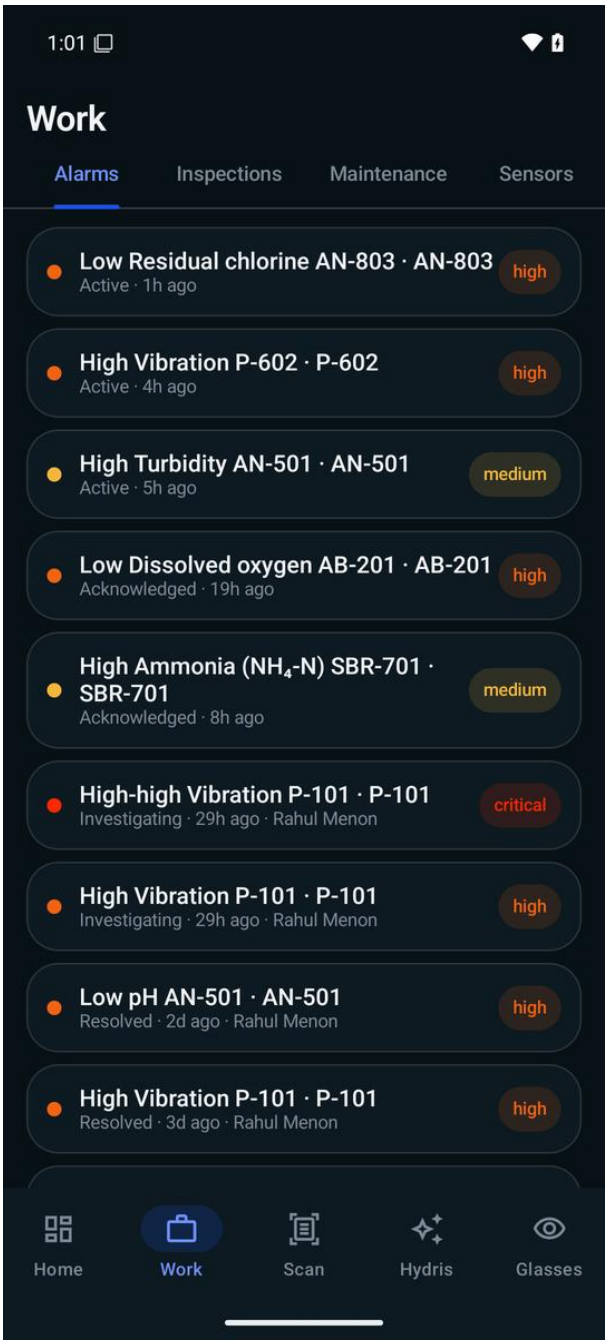


Figure 7. *Android work list with alarms grouped by severity*

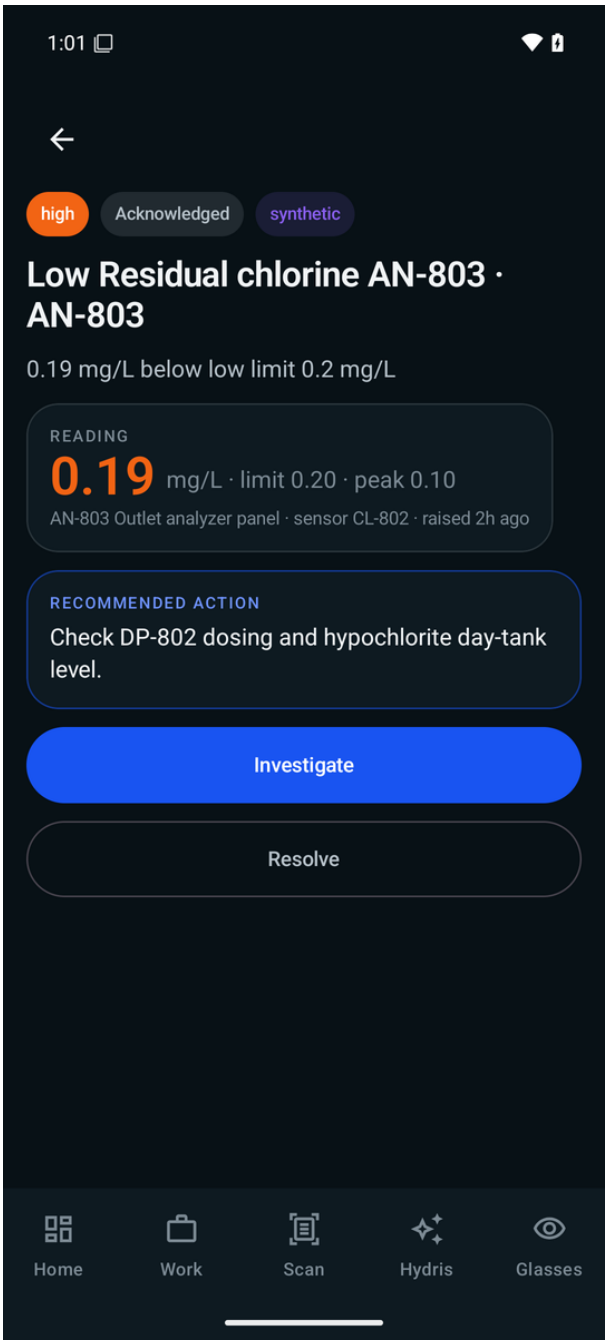


Figure 8. *Android alarm detail: reading, limit, recommended action and the next lifecycle action*

7.9 Charts

Web charts (`apps/web/src/ui/charts.tsx`) are hand-built SVG microcharts, plus `TrendChart` built on Recharts `AreaChart`. Every chart reads colours from CSS variables, is an accessible image (`role="img"` with `aria-label`), can render a screen-reader data table (`SrTable`), and shows an explicit empty state instead of inventing numbers. Android charts are drawn on Compose `Canvas` without a chart library (`Components.kt`; the mobile spec's Vico choice was not adopted).

Table 22. Chart specifications

Chart	Web	Android	Specification
Sparkline	`Sparkline`: default 40 px tall, stroke `--color-lime`	`Sparkline`: 2 dp round stroke, range padded 7.5 % top and bottom	No axes; trend only; always next to the current value

Chart	Web	Android	Specification
Line / trend	`TrendChart` (Recharts area, 260 px, thresholds as reference lines, typical band); `MiniTrend` with band and limits	`LineChart`: typical range shaded in `OkSoft` 8 %, 3 gridlines, 28 % gradient fill, end-point dot	Thresholds and typical range always drawn when known
Bars	`Sparkbars` (peak emphasis), `BarStrip`, `BarList`, `StackedBar`	`Bars`: 4 dp gaps, 4 dp radius, last bar full colour, others 45 %	Highlight only the bar that matters (latest or peak)
Donut	`Donut`: 156 px, 14 px ring, centre value, legend with percentages	`Donut`: 14 dp stroke, 2 degree gaps, starts at 12 o'clock, centre slot	Compositions only; at most 5 segments
Gauges	`Gauge`, `RadialGauge` with marks and band	n/a	Single value against a range
Heatmap	`Heatmap` (week by hour)	n/a	Event density
Progress	`Progress`, `BulletBar` (target)	`ProgressBar`	Fraction of a known total

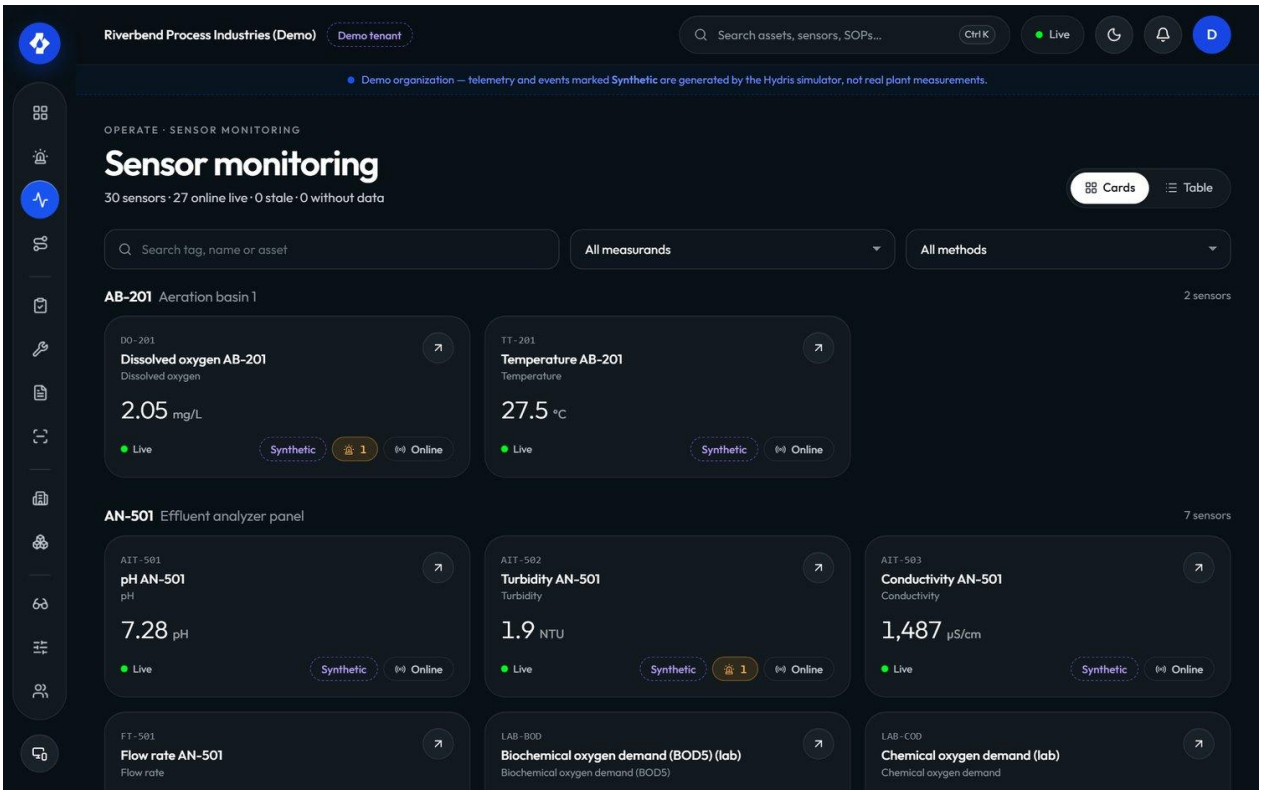


Figure 9. *Console telemetry: trend chart with thresholds and freshness*

7.10 Empty, loading and error states

Table 23. Data state patterns

State	Web	Android	HUD
Loading	`LoadingBlock` skeleton rows with shimmer, `role="status"`; `QueryView` keeps last good data while refetching	`Loading`: Signal Blue spinner	"Generating…", "Context will load when the connection returns"
Empty	`EmptyState`: dashed border, concentric art, title, body, action	Inline text ("No …"), per screen	Plain 22 unit sentence ("This alert is no longer open.")

State	Web	Android	HUD
Not found	`ErrorState` turns 404 and invalid ids into "We couldn't find that record" with a back link	n/a	n/a
Forbidden	"You do not have access to this" with admin guidance	Server error text	`NotPermitted` screen and `Blocked` notes
Error	Red-tinted `role="alert"` panel with message and Retry	`ErrorNote`: High tint panel with message	Banner (`role="status"`) in critical or high tone
Offline	Realtime status in shell	Error notes	Offline strip plus queued counter

8 Data visualisation guidelines

1. **Show the value, its unit and its time together.** A number without freshness is not shown. Use `FreshnessBadge` (web) or `Hx.freshness()` (Android: live `OkSoft`, delayed `Medium`, stale `High`, no data `Muted`).
2. **Never plot invented data.** Missing readings are gaps, not zeros (`freshness.ts`); charts render their empty state ("No data", "No records").
3. **Limits and typical ranges are context, not decoration.** Draw thresholds as reference lines and typical ranges as a faint band (Android `LineChart`, web `MiniTrend` and `TrendChart`).
4. **One highlight per chart.** Emphasise the latest or peak bar; mute the rest (`Sparkbars` `emphasise`, Android `Bars` 45 % alpha).
5. **Severity colours only for severity.** Series colours come from the brand set (Signal, Core Blue, Insight Purple, text-2) in that order; Alert Orange is reserved for critical.
6. **Every chart has a text equivalent.** Web: `aria-label` plus optional `SrTable`. Android: **Planned** (`Modifier.semantics { contentDescription }` on chart canvases).
7. **Tabular figures** for every numeric axis, legend and value.
8. **HUD: no charts beyond a single sparkline.** On the glasses, a reading and its state word beat any chart.

9 Layout grids

9.1 Console desktop

The console shell (`apps/web/src/console/Shell.tsx`) has a fixed 84 px icon rail on the left from the `md` breakpoint, a sticky blurred header with the search trigger (Ctrl or Cmd K command palette), theme toggle and account menu, an optional demo banner, and a main column capped at 1440 px with 32 px gutters. Below `md` the rail becomes a floating bottom capsule bar with four primary items (Overview, Alarms, Sensors, Inspections) and More, which opens an "All modules" bottom sheet.

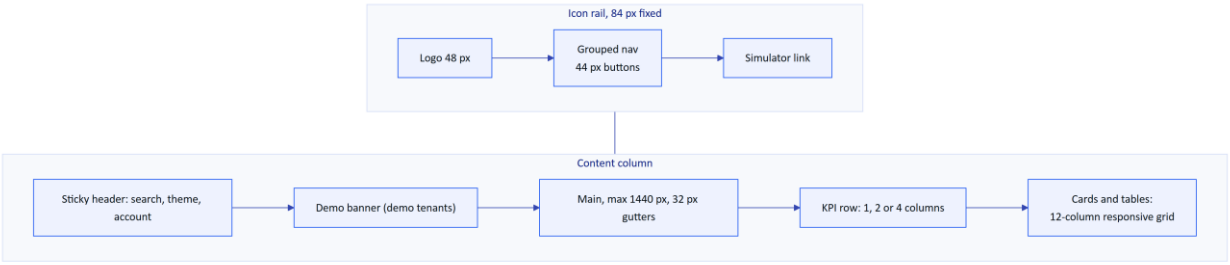


Figure 10. *Console desktop layout regions (md breakpoint and wider)*

Table 24. Console breakpoints and grid behaviour

Breakpoint (Tailwind)	Navigation	KPI row	Content grid
Below `sm` (640 px)	Bottom capsule bar plus More sheet	1 to 2 columns	Single column; dialogs become bottom sheets
`sm` to `md`	Bottom capsule bar	2 columns	Single column; centred dialogs
`md` (768 px) and up	84 px rail	2 columns	Two-column card layouts begin
`lg` (1024 px)	Rail	2 to 3 columns	Main plus side column
`xl` (1280 px) and up	Rail	4 columns (`xl:grid-cols-4`)	Full dashboard grid, max 1440 px

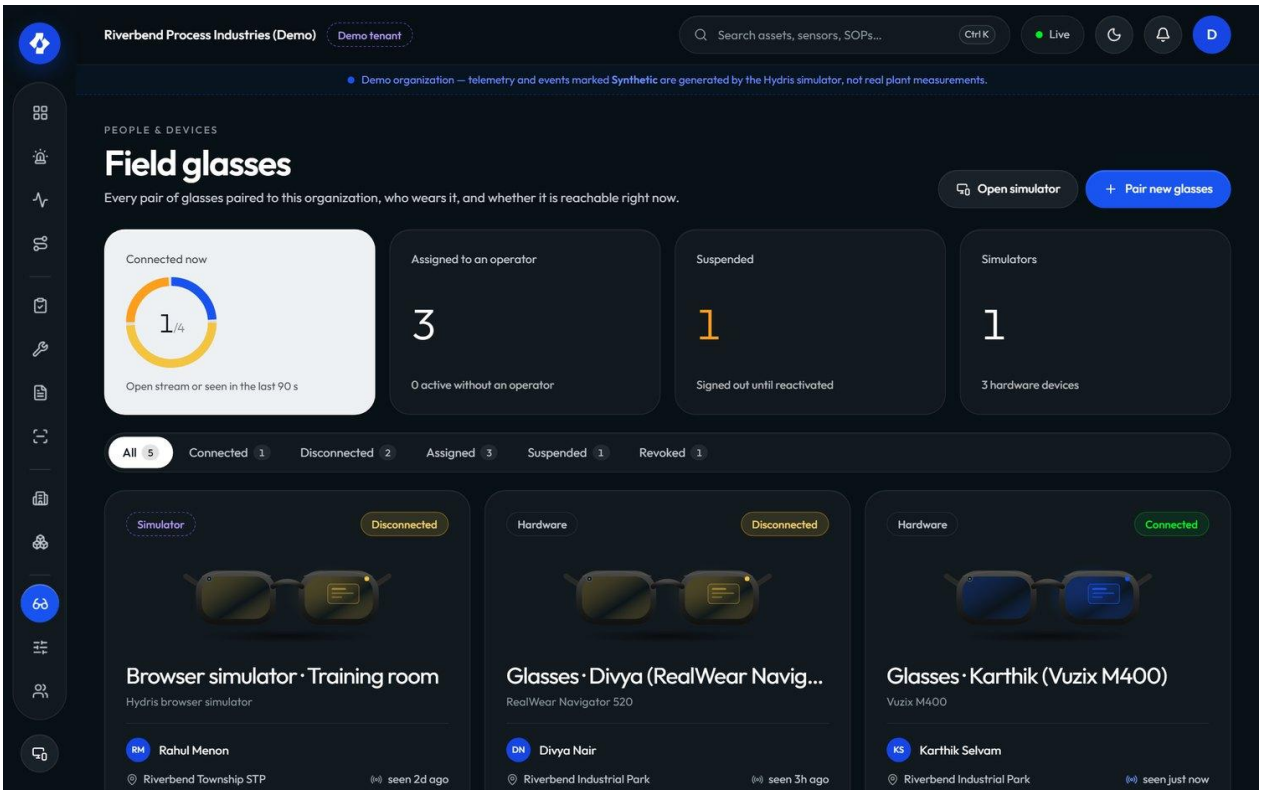


Figure 11. *Console devices: table layout inside the 1440 px content column*

9.2 Phone (glasses phone mode and Android)

The glasses web app switches to **phone mode** for portrait phones, the Ray-Ban Meta companion app and any page with `?mode=phone` (`useViewportLayout` in `hud.tsx`). Phone mode keeps a 540-unit design width (`PHONE_DESIGN_WIDTH`), scales by `window width / 540`, and grows to the full device height so screens fill the phone instead of sitting in a letterboxed square. Safe areas are respected with `env(safe-area-inset-top)` on the status bar and `env(safe-area-inset-bottom)` on the tab bar.

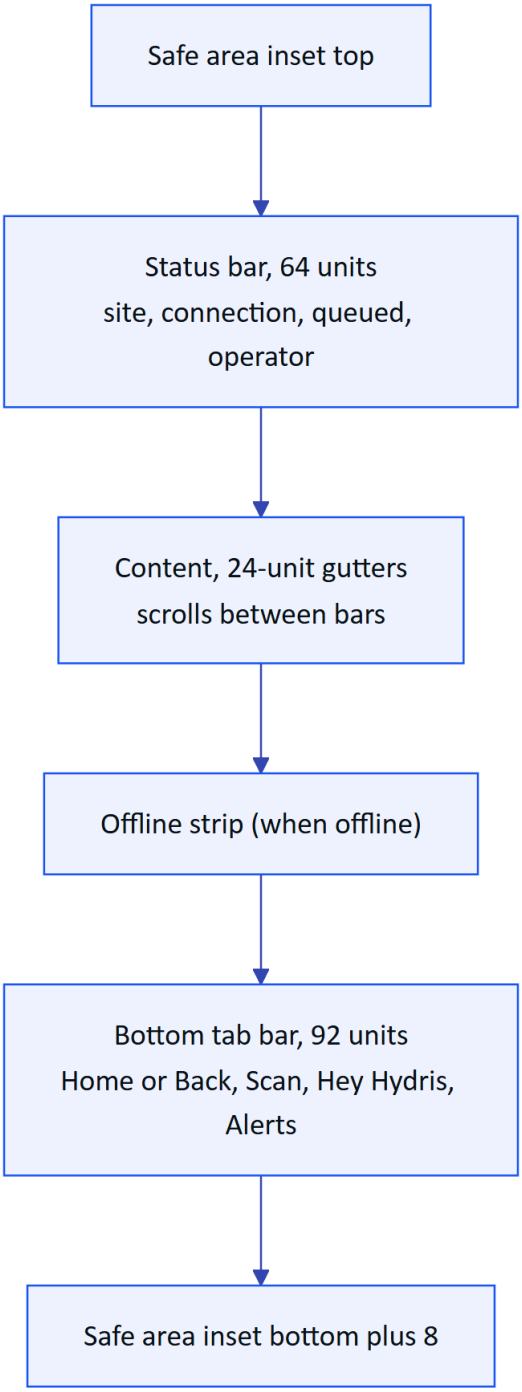


Figure 12. *Phone mode layout, 540-unit design width with safe areas*

The Android app follows the same structure natively: `statusBarsPadding()`, 16 dp horizontal padding, a `LazyColumn` of cards with 12 dp spacing, and a Material `NavigationBar` with five tabs (Home, Work, Scan, Hydris, Glasses) on `Hx.Surface`, selected items in `SignalText` with an 18 % Signal indicator (`ui/Root.kt`).

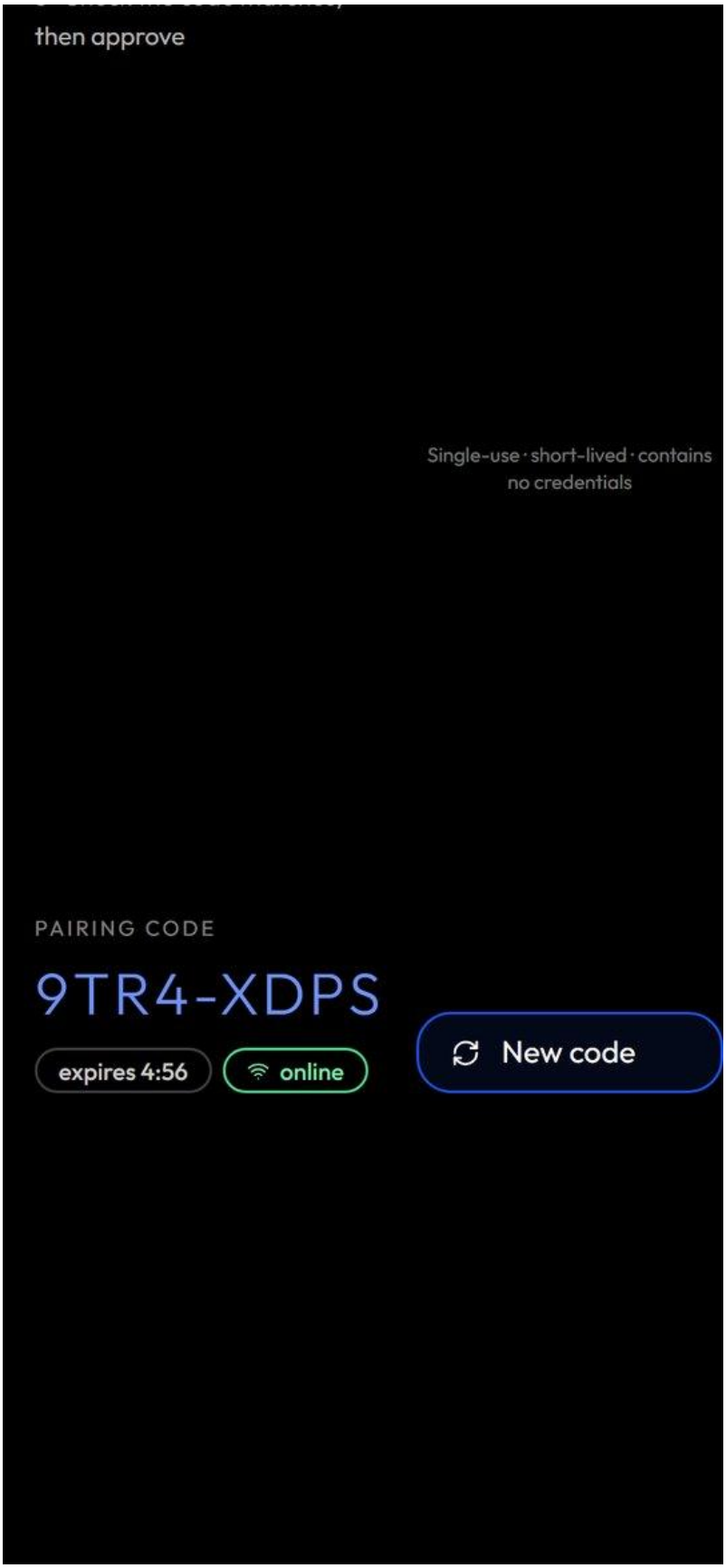


Figure 13. *Glasses app in phone mode: pairing screen on a phone*

9.3 HUD 600 by 600

The HUD canvas is 600 by 600 units scaled to fit (`useFitScale`, `useViewportLayout`). The simulator also offers 640 by 480 and 480 by 480 viewports (`apps/web/src/app/simulator/page.tsx`) to test other display shapes.

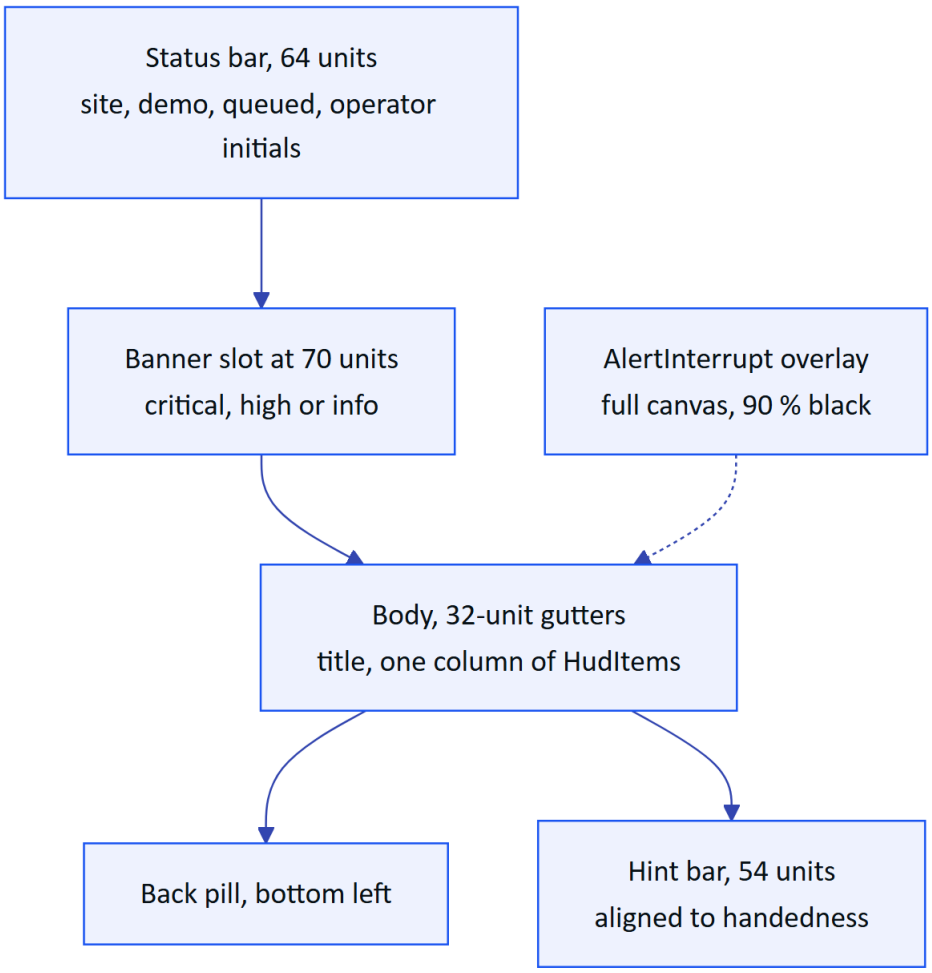


Figure 14. *HUD 600 by 600 layout regions*

Table 25. **HUD layout rules**

Rule	Value	Source
Canvas	600 by 600 units, black (renders transparent)	`hud.tsx`
Gutters	32 units (HUD), 24 units (phone)	`GlassesApp.tsx`
Columns	One	`hud.tsx` header comment
Focus	One focused `HudItem` with Signal border and 10 % Signal fill	`HudItem`
Row height	14 units vertical padding (comfortable), 9 units (compact)	`HudItem`, `layout.density`
Primary actions	One per screen	`hud.tsx` header comment
Hint bar	Right or left aligned per `layout.handedness`	`HintBar`
Text scale	`layout.textScale` 0.85 to 1.4	`uiprofile.ts`

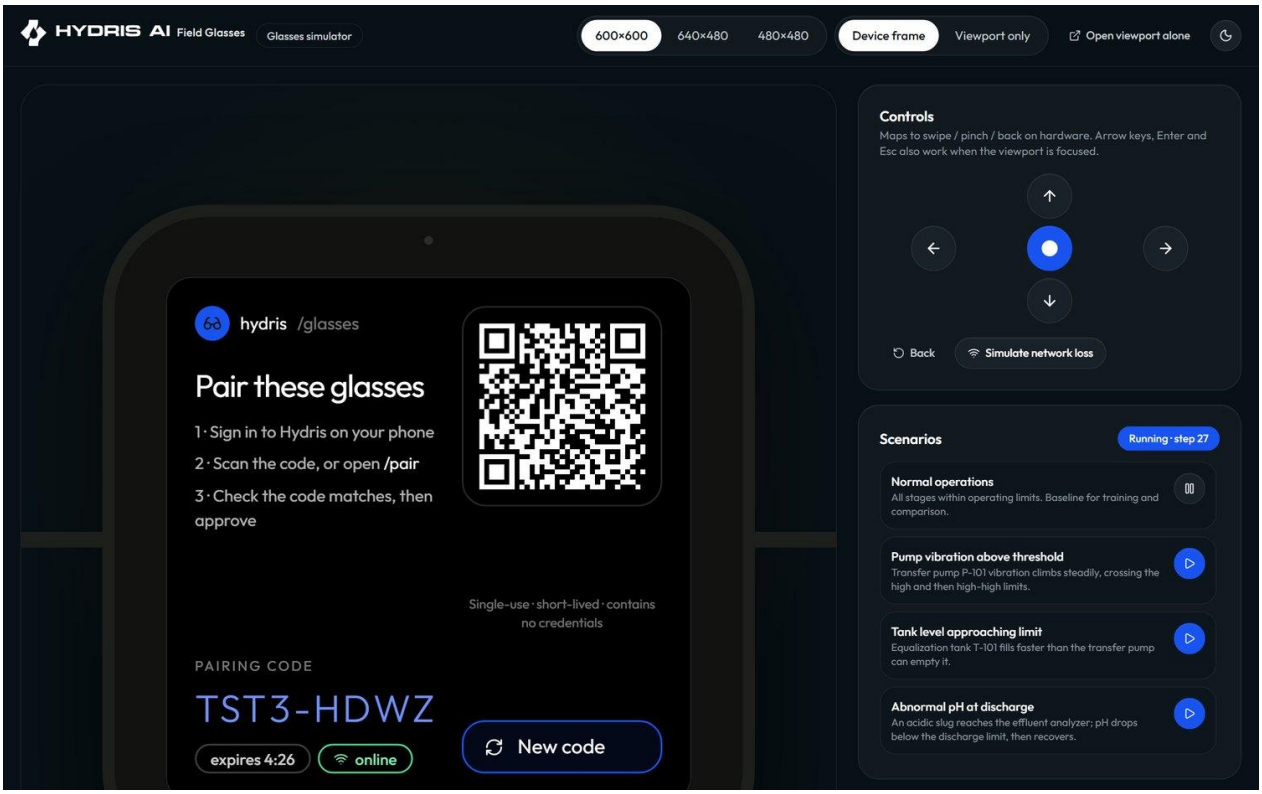


Figure 15. *Glasses simulator: HUD viewport, device frame, input pad and event console*

9.4 UI profiles shape the layout

Administrators control what the glasses show through versioned UI profiles (`packages/domain/src/uiprofile.ts`): 12 screens (Home and Device and session can never be disabled), 4 home widgets, 8 actions, a measurement allow-list and per-asset-type order, plus layout density, text scale and handedness. The most specific published assignment wins entirely: device, then user, then role, then organisation, then the built-in default. Designers must design every glasses screen for both densities, all four text scales and both hands.

10 Navigation patterns

10.1 Console information architecture

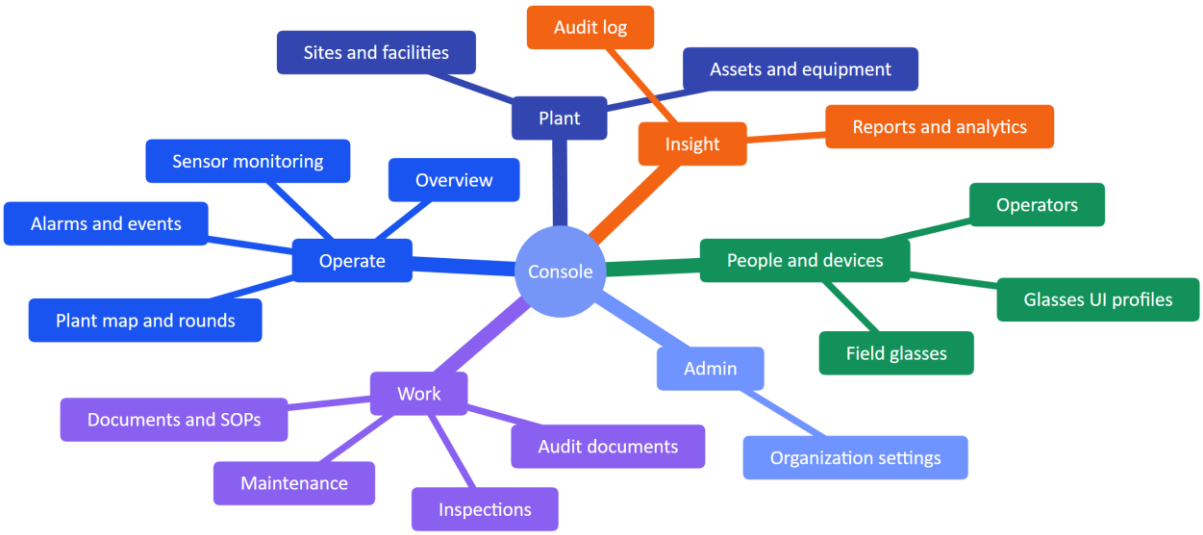


Figure 16. *Console information architecture (navigation groups in Shell.tsx)*

Navigation items are filtered by permission (`perm` on each `NavItem`), so an operator never sees a module they cannot open. The command palette searches assets, sensors, documents and people from every page.

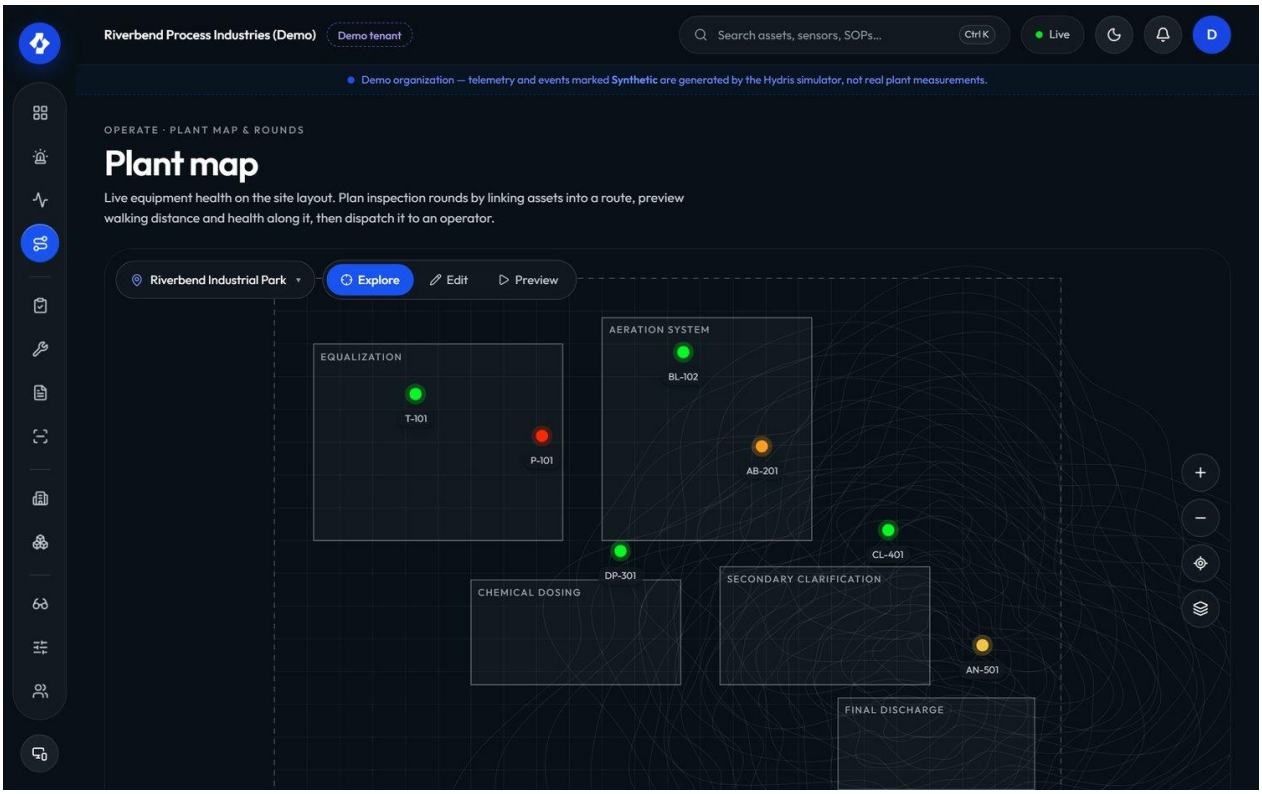


Figure 17. *Console plant map and rounds*

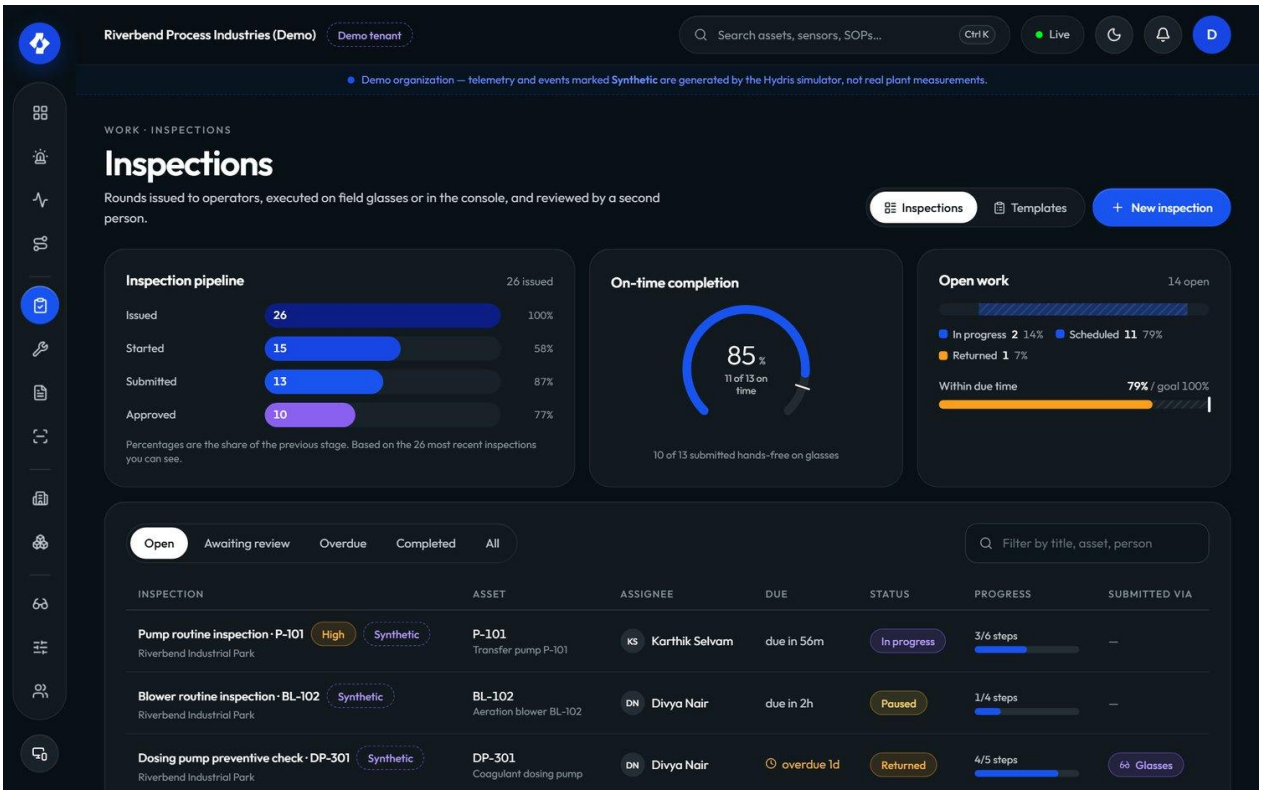


Figure 18. *Console inspections*

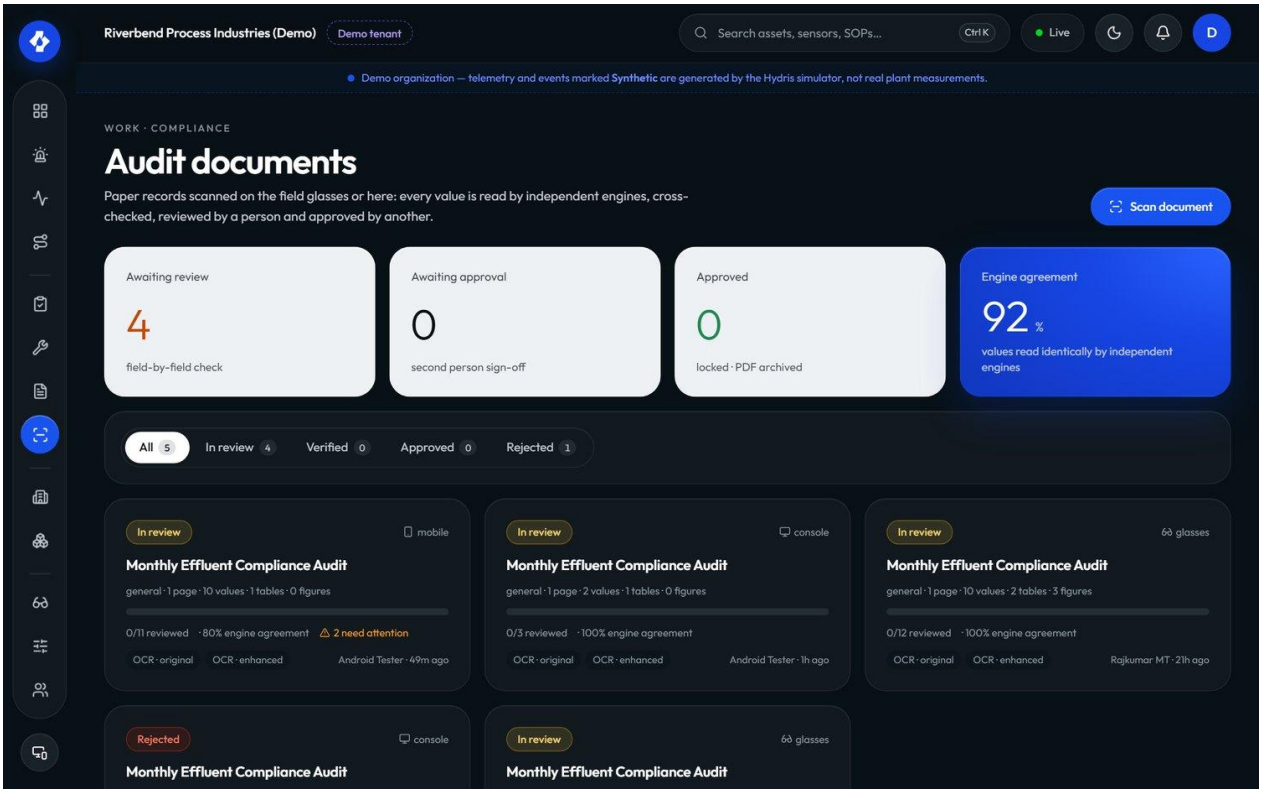


Figure 19. *Console audit documents: review and approval of scanned records*

10.2 Glasses app information architecture

The glasses app is a stack: Home is the root, every selection pushes a screen, and Back pops. Screens are gated by the UI profile (`screenAllowed` logic at the top of `GlassesApp.tsx`).

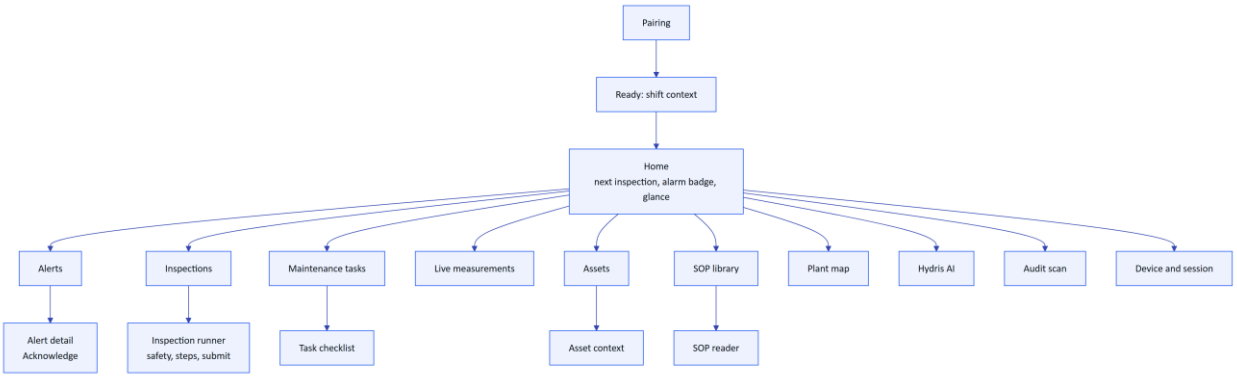


Figure 20. *Glasses app navigation stack (profile-gated screens)*

10.3 Android app navigation

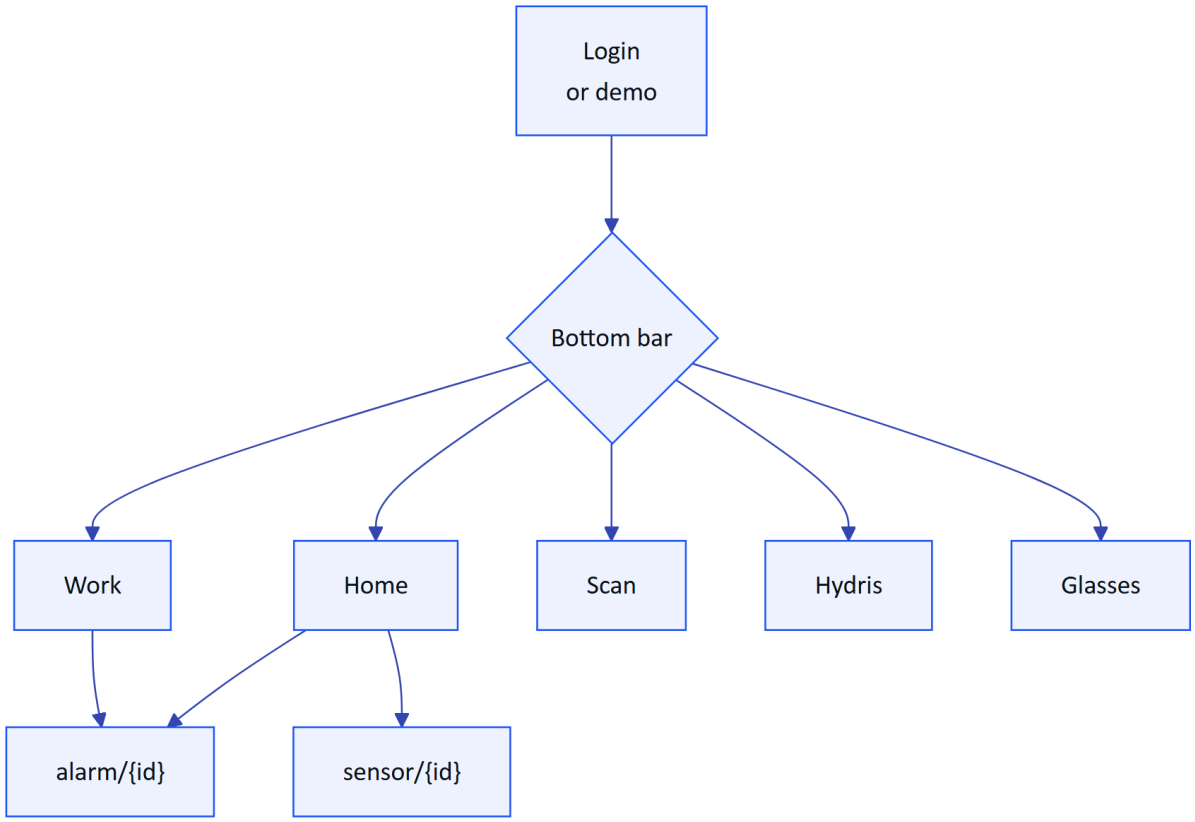


Figure 21. *Android app navigation (ui/Root.kt)*

Tab switches pop to Home with saved state and `launchSingleTop`, so returning to a tab restores its scroll position. Detail screens show a back arrow at top left (`Icons.AutoMirrored.Outlined.ArrowBack`), and the system back gesture works everywhere.

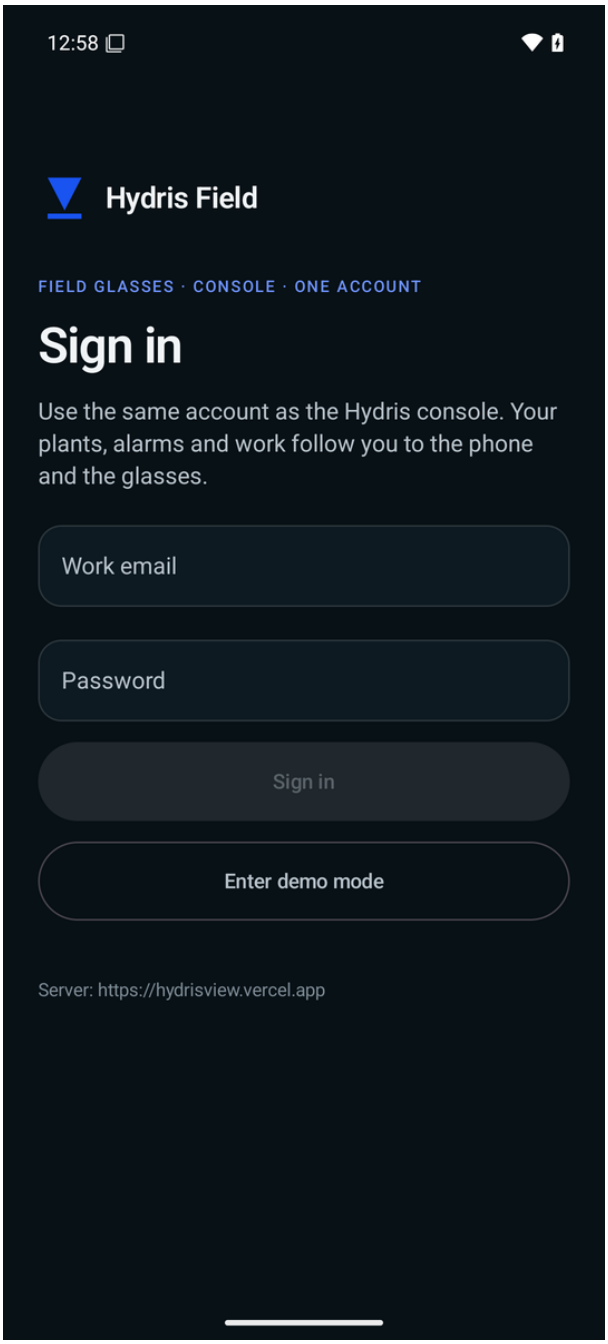


Figure 22. *Android login*

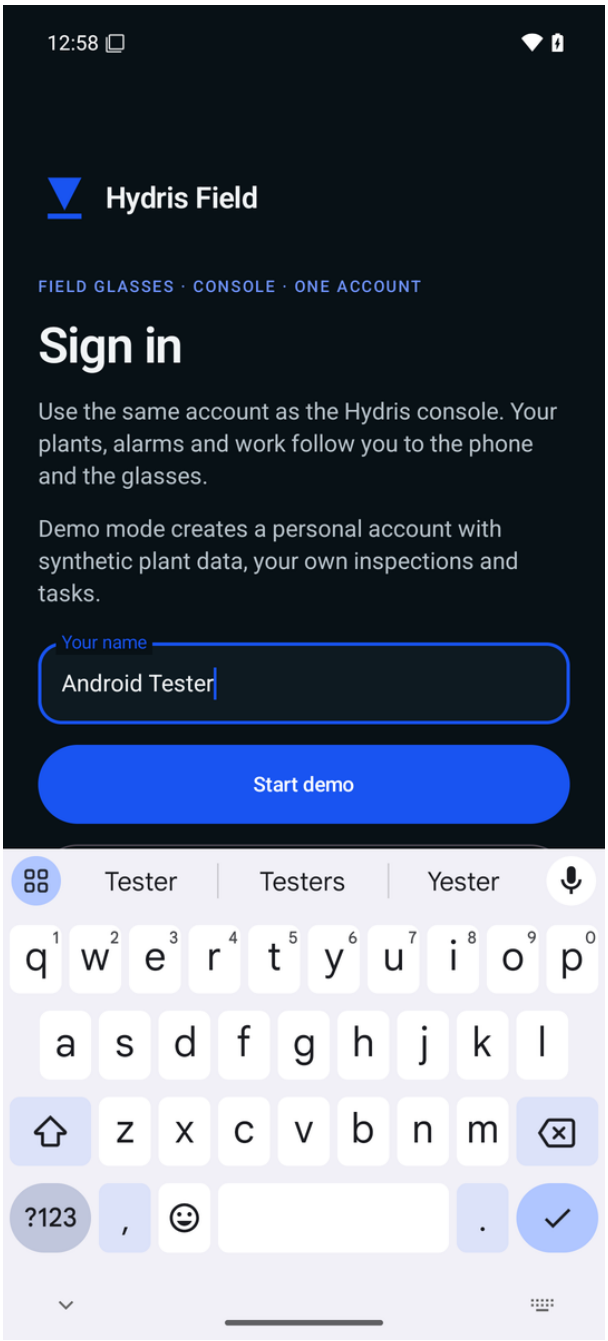


Figure 23. *Android demo sign-in: choose a display name*

11 Interaction patterns

11.1 Input model

The glasses app reduces every input to six keys: up, down, left, right, select, back (`apps/web/src/glasses/input.ts`). Keyboard arrows, Enter or Space, Escape or Backspace, touch swipes of at least 40 px, and simulator `postMessage` controls all map onto that vocabulary. Hardware gestures (neural band pinch and swipe, frame touchpad) plug in once their delivery to the web runtime is verified on a device (**Planned**). Phone mode adds direct touch on rows and the bottom tab bar.

Table 26. Input vocabulary and its meaning

Key	HUD meaning	Phone mode	Android
Up / Down	Move focus between `HudItem`s; change a digit in numeric entry	Swipe up or down	Scroll
Left / Right	Move between digits; right also opens	Swipe right to open, left for back	n/a
Select	Activate focused item	Tap	Tap
Back	Pop the stack; dismiss interrupt	Back tab or swipe left	Back arrow or system back
Voice	"Hey Hydris"	Hey Hydris tab	Mic toggle on Hydris tab
Capture button	n/a on display glasses	n/a	Ray-Ban Meta short press triggers a scan capture

11.2 Voice first: "Hey Hydris"

The wake phrase is matched by `WAKE_RE` in `packages/domain/src/voice.ts`, which tolerates common mishearings ("hey hydras", "ok hydrus", "hi dris"). A question can follow in the same breath ("Hey Hydris, what's the pH?"). Answers come from the account's live data through `POST /assistant/ask`, are shown as a card and spoken through the glasses speakers or the device's speech synthesis (`VoiceAssistant.tsx`).

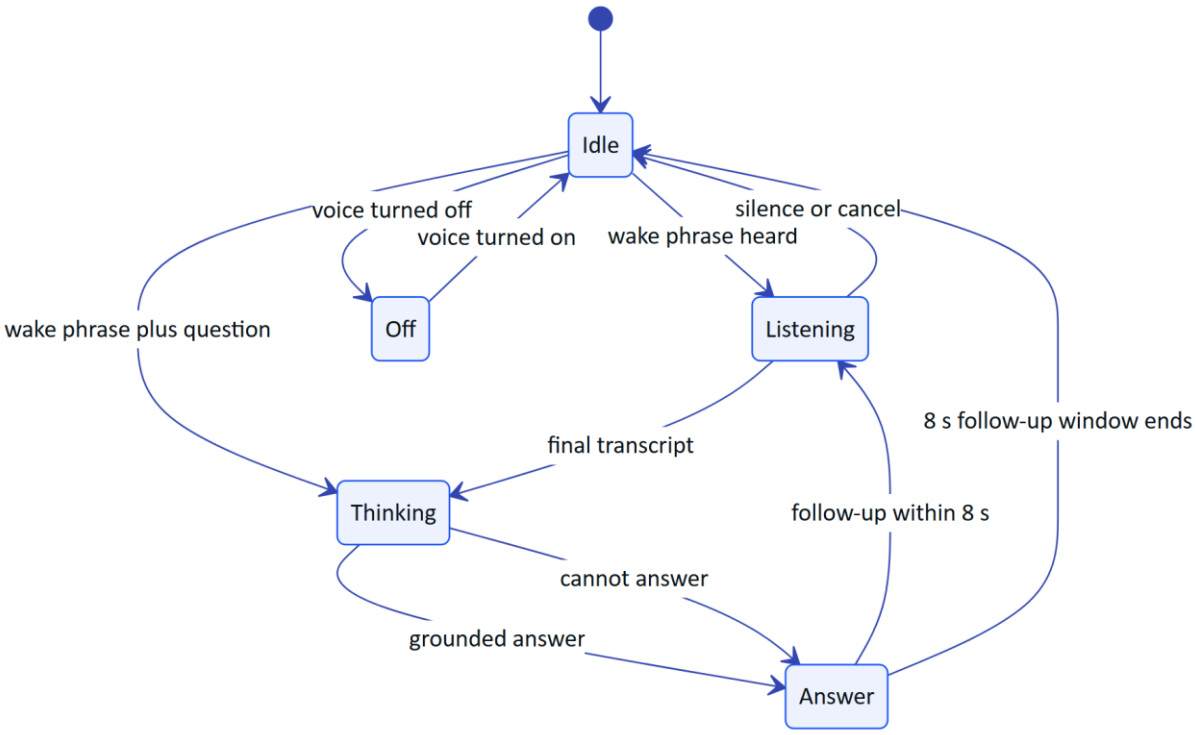


Figure 24. *Hey Hydris voice states (VoiceAssistant.tsx)*

Design rules for voice:

- Always show what was heard (Android renders the transcript in `SignalText` quotes) before the answer arrives.
- The thinking state has copy ("Checking your plant data…") so silence is never ambiguous.
- Answers list their facts with value, unit and label, so the operator can check the source.
- The mic state is always visible: "Hey Hydris" (listening for wake), "Voice off", "No mic" (`VoiceAssistant.tsx`).
- Voice never performs a state change on its own. Acknowledging or submitting stays an explicit select or tap.

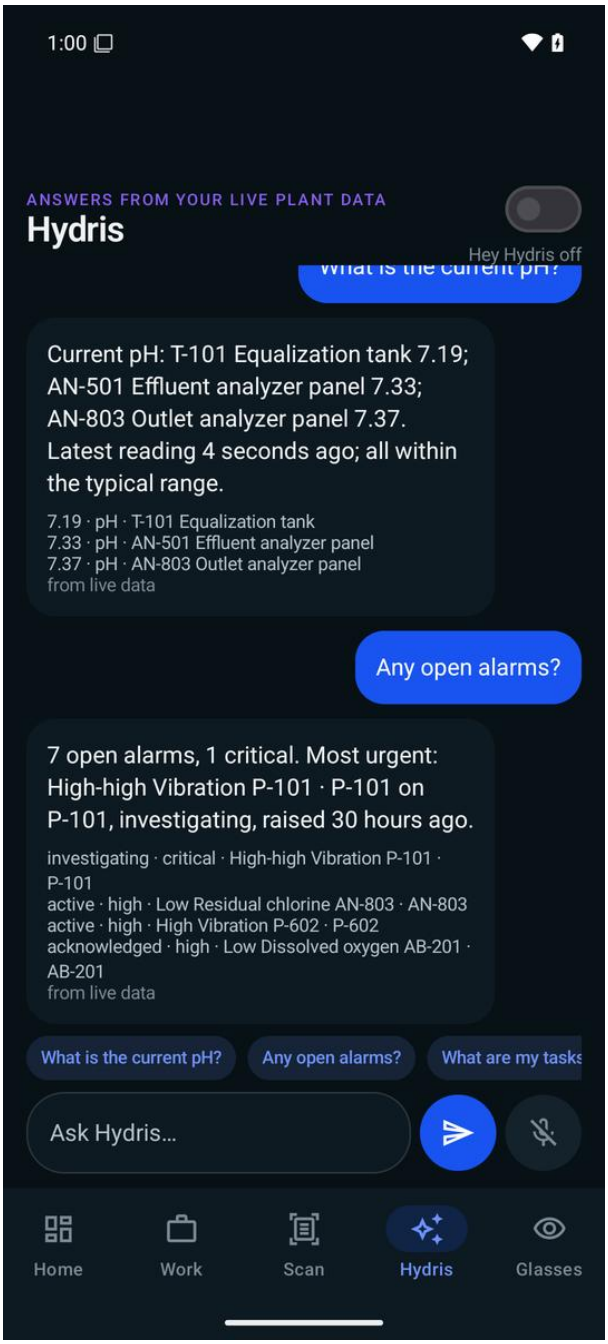


Figure 25. *Android Hydris assistant: grounded answer with facts*

11.3 Capture button and audit scan

On Ray-Ban Meta glasses the physical capture button is surfaced through the Meta Wearables Device Access Toolkit: a `SHORT_PRESS` capture event is emitted on `GlassesManager.captureButton` and the scan screen uses it to capture a page (`apps/android/.../glasses/GlassesManager.kt`). The scan menu offers three sources: glasses camera, phone camera with guided A4 framing, and photo import (`ScanScreen.kt`).

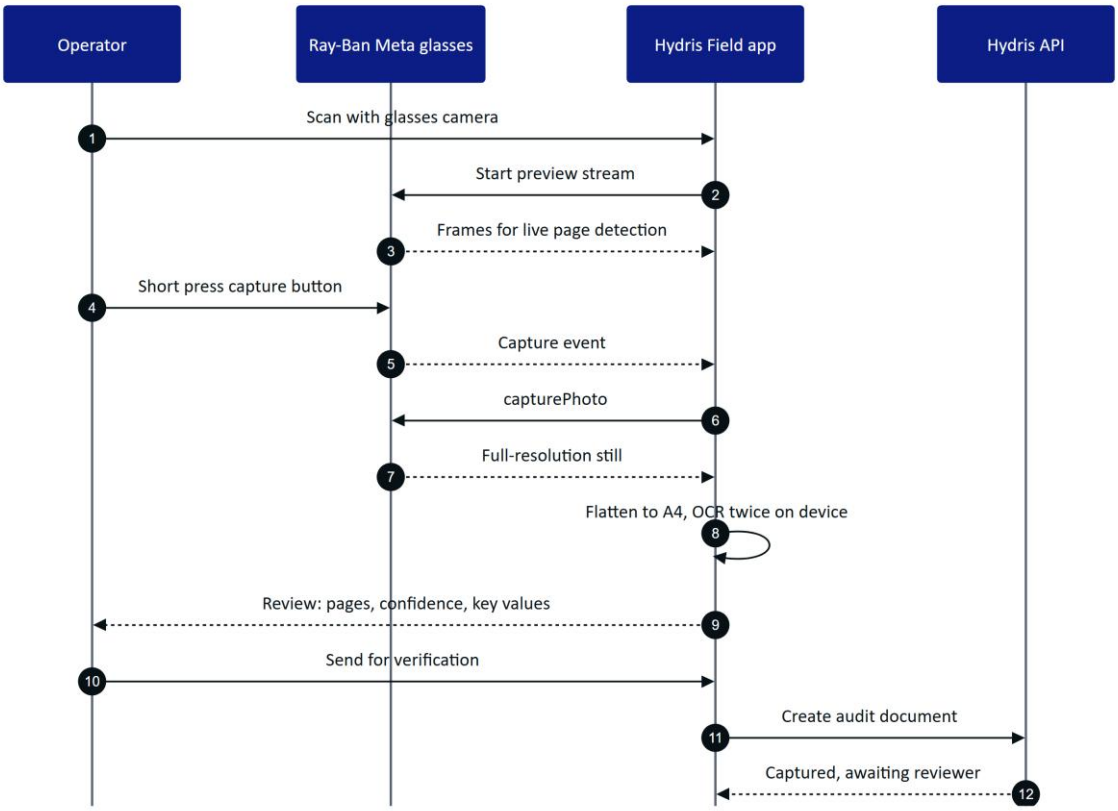


Figure 26. *Audit scan with the glasses capture button*

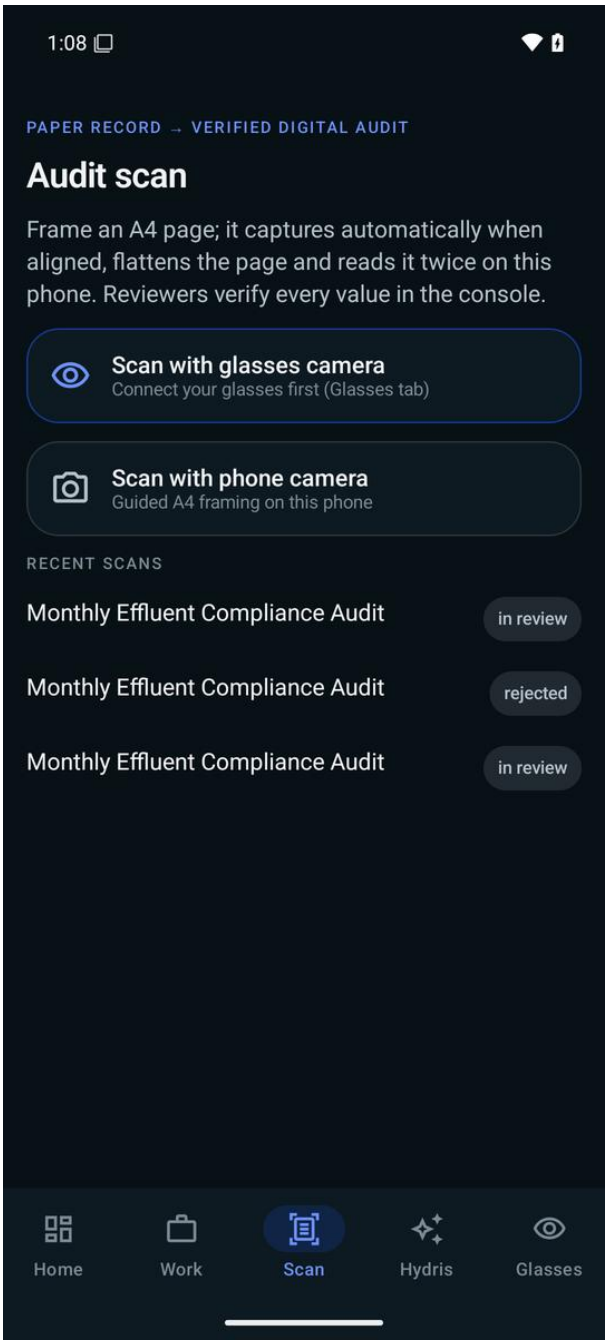


Figure 27. *Android scan menu: glasses camera, phone camera or import*

11.4 Guided capture feedback

`GuidedCapture` (`apps/web/src/lib/GuidedCapture.tsx`, with logic in `lib/docdetect.ts`; ported to Kotlin in `apps/android/.../scan/DocDetect.kt`) shows one instruction at a time. Detection runs every 90 ms; the detected sheet is outlined in Signal Blue while guidance is pending, in orange `#FF6A3D` when it touches the edge or is skewed, and in Active Green once ready. After 900 ms of continuous readiness it captures automatically.

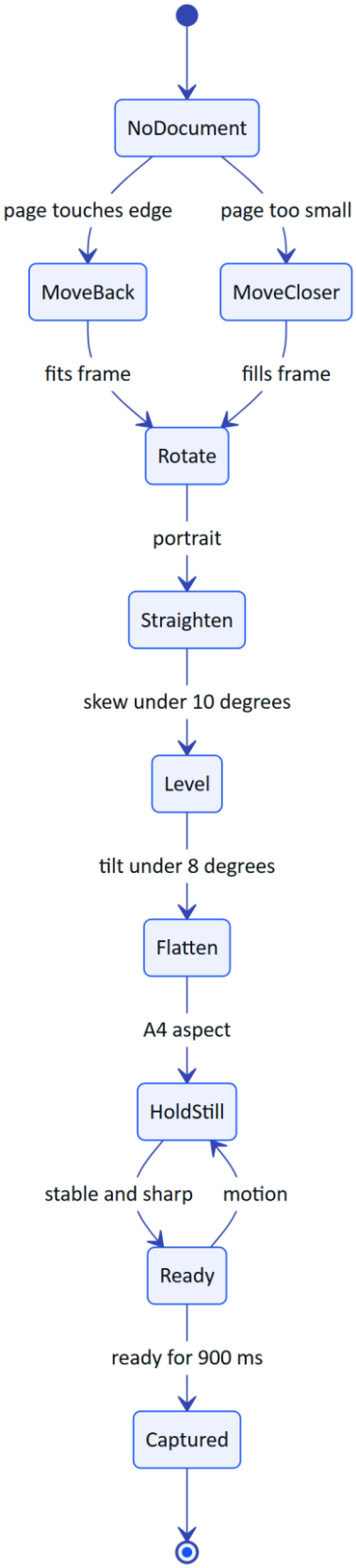


Figure 28. *Guided A4 capture guidance (docdetect.ts guide function)*

Table 27. Guidance messages (lib/docdetect.ts)

Code	Message shown	Outline colour
`no_document`	Place the A4 sheet in view, on a dark surface	None
`move_back`	Move back - show all four corners of the page	Orange
`move_closer`	Move closer - fill the frame with the page	Signal Blue
`rotate`	Turn the sheet to portrait (long side vertical)	Signal Blue
`straighten`	Look straight down at the page - reduce the tilt	Orange
`level`	Level the page - rotate it left or right until upright	Orange
`flatten`	Hold the sheet flat and square to your view	Signal Blue
`hold_still`	Hold still…	Signal Blue
`ready`	Perfect - capturing	Active Green, with fill progress

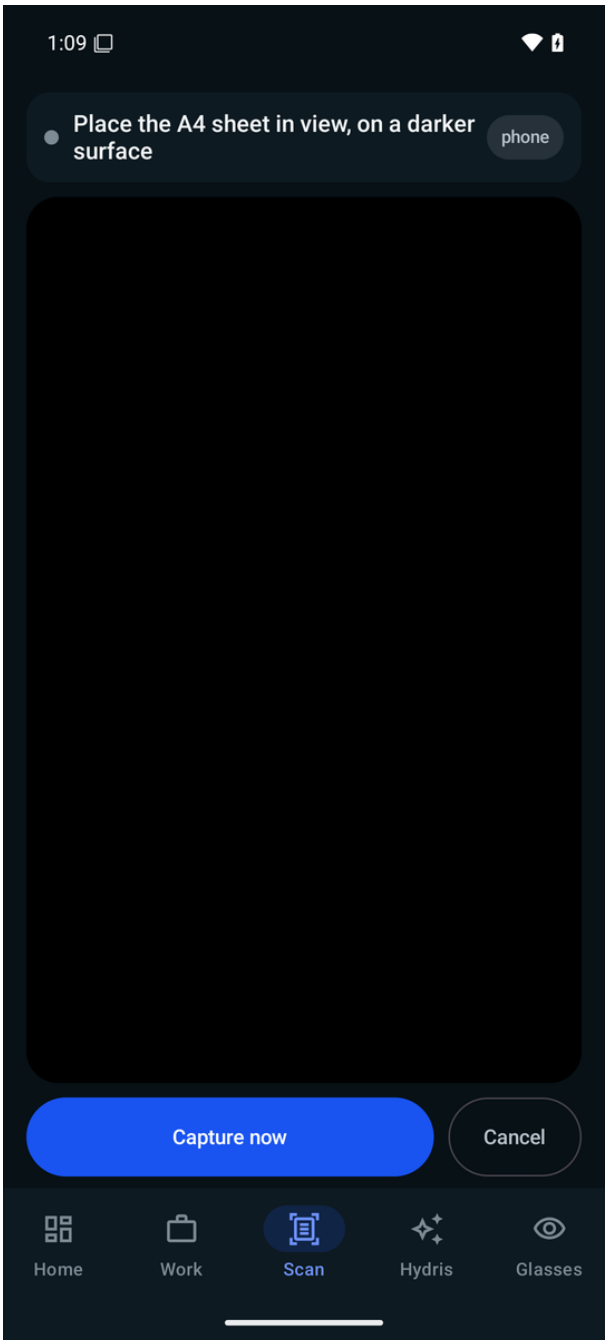


Figure 29. *Android guided capture: live outline and single instruction*

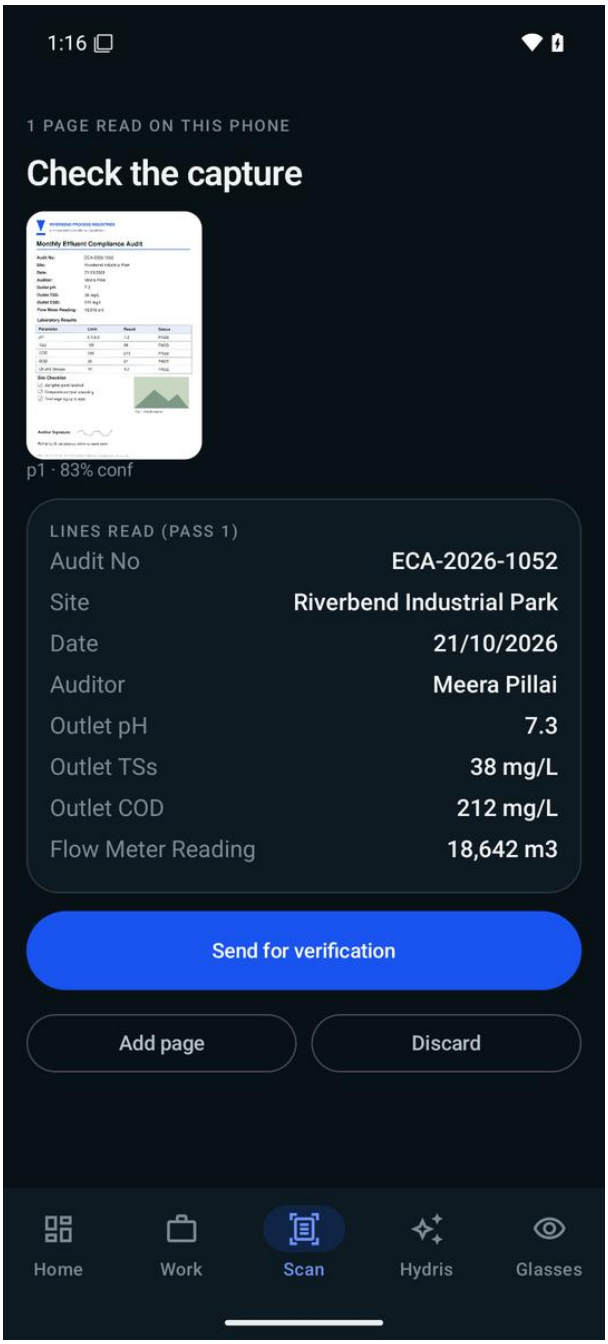


Figure 30. *Android scan review: pages, confidence and detected values*

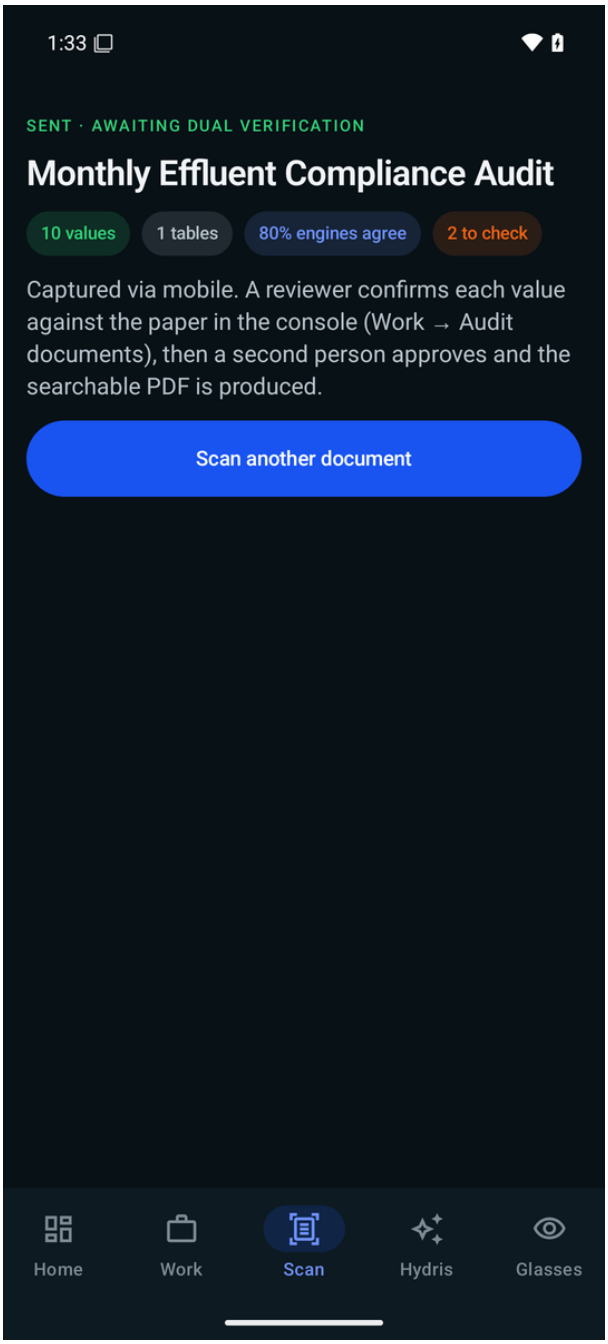


Figure 31. *Android scan sent: next steps for reviewer and approver*

11.5 Back navigation

- HUD: Back key or swipe left pops one screen; a Back pill sits bottom left when the stack is deeper than Home.
- Phone mode: the first tab becomes **Back** whenever the stack is deeper than Home, otherwise **Home** (`PhoneTabBar`). This keeps back reachable with a thumb at the bottom of the screen.
- Android: top-left back arrow on detail screens and the system back gesture.
- Console: breadcrumbs and `BackLink`; `ErrorState` defaults back to the parent list.
- Back never discards recorded work: inspection steps are saved as they are recorded and queued offline.

11.6 Alarm acknowledgement and lifecycle

The alarm lifecycle is defined once in `packages/domain/src/alarms.ts` and enforced by the API. Each surface shows only the next valid actions.

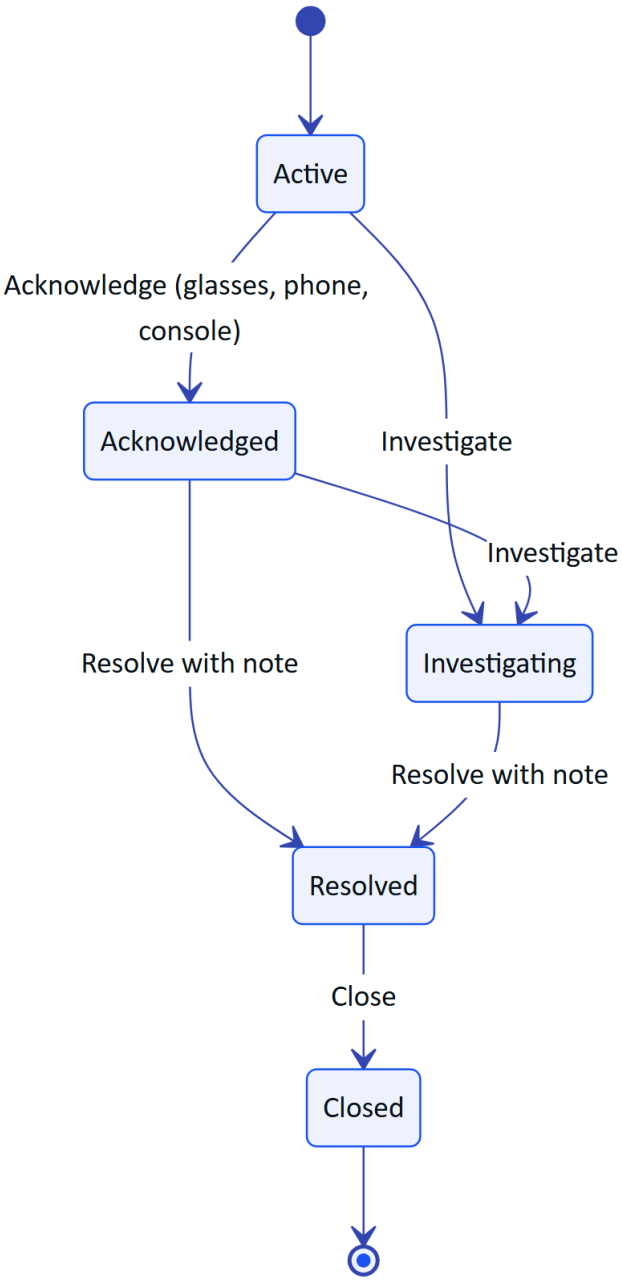


Figure 32. *Alarm lifecycle and the actions each surface offers*

Table 28. Acknowledgement and lifecycle patterns by surface

Surface	Interrupt	Acknowledge	Resolve	Status
Glasses HUD	`AlertInterrupt` full-screen overlay for critical and high, View or Dismiss	Single focused `HudItem`, disabled offline with "(needs connection)"	Not offered (console or phone)	**Built**
Android	None (push notifications **Planned**)	Full-width primary button	Button shown, but no note is collected	**Partial**

Surface	Interrupt	Acknowledge	Resolve	Status
Console	Realtime toast for critical alarms	Button on detail	`ConfirmDialog` with required note "What was done to resolve the condition"	**Built**

> **WARNING**
>
> On Android the Resolve button posts the action without a note (`WorkScreen.kt` calls `alarmAction(id, "resolve")`), but the API rejects resolve without one (`note_required` in `apps/api/src/modules/alarms.ts`). The operator sees an error note. Fix: a bottom sheet that collects the note, matching the console's `ConfirmDialog`. Status: **Planned**.

11.7 Operator experience journey

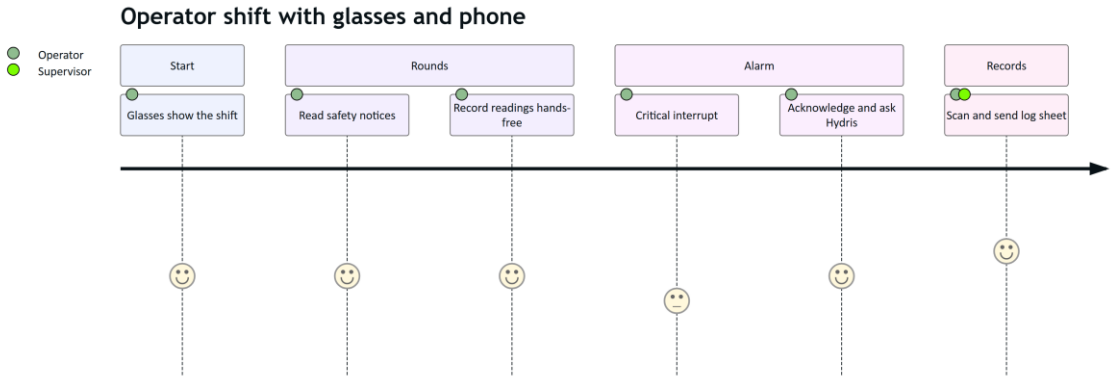


Figure 33. *Field operator shift journey with Hydris (target experience)*

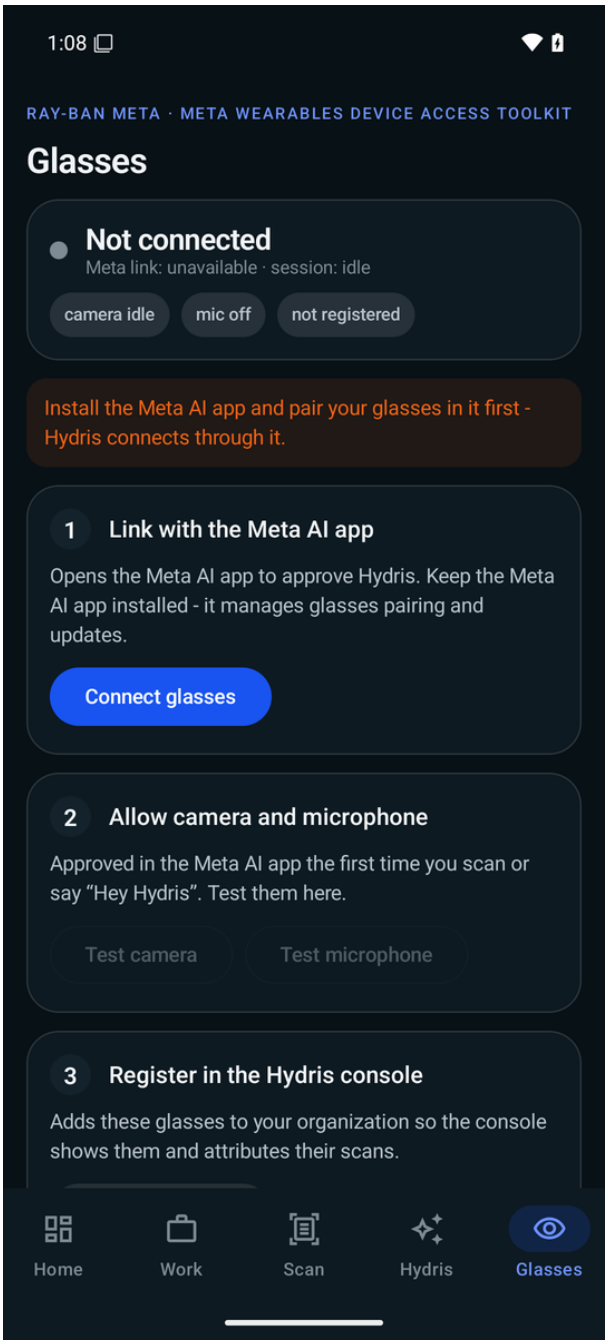


Figure 34. *Android glasses screen: connection checklist and registration*

12 Motion

Motion explains change; it never decorates. All motion tokens are in globals.css.

Table 29. Motion tokens and patterns

Token or class	Value	Use
`--speed-fast`	150 ms	Colour and border changes, press feedback
`--speed`	240 ms	Lifts, fades
`--ease-out`	cubic-bezier(0.22, 1, 0.36, 1)	Default easing
`--ease-spring`	cubic-bezier(0.34, 1.4, 0.64, 1)	Theme toggle icon only
`.page-enter`	360 ms fade and 8 px rise	Route change in console
`.pop-in`	220 ms	Popovers, centred dialogs
`.sheet-up`	320 ms	Bottom sheets
`.slide-in-right`	320 ms	Side panels
`.stagger`	40 ms steps up to 200 ms	Card grids on load
`.pulse-dot`	1.8 s	Live freshness dot only
`.skeleton`	1.6 s shimmer	Loading
Chart transitions	600 to 700 ms	Bars and arcs growing to value
Theme switch	300 ms colour cross-fade	`html.theme-switching`

Rules:

- `prefers-reduced-motion: reduce` disables animations and transitions globally, stops the pulse dot and hides the toast timer (globals.css). **Built** on web.
- The HUD uses colour transitions only; no movement on an additive display, where motion pulls the eye from the task. **Built.**
- Android uses Material defaults; honouring the system "remove animations" setting is inherent to Compose animation APIs. Custom motion is not used. **Built.**
- Alarms never animate in a way that could be mistaken for normal motion; the interrupt simply appears.

13 Accessibility

13.1 Targets

Table 30. Accessibility targets

Area	Target	Status
Text contrast	WCAG 2.2 AA (4.5:1, 3:1 large) on every surface and theme	**Partial** (3 failures listed in the contrast audit)
Non-text contrast	3:1 for borders of inputs, focus rings, chart marks	**Built** (focus ring `#6F93FF` 6.62:1 on bg)
Target size	WCAG 2.5.8: at least 24 by 24 px; Hydris standard 44 by 44 px	**Built** on web (`::after` hit areas on `btn-sm`, `icon-btn-sm`, chips, links) and Android (48 dp Material, 52 dp buttons)
Focus visible	2 px outline, 2 px offset, colour follows surface	**Built** (`:focus-visible` in globals.css)
Keyboard	Every console action reachable by keyboard; HUD fully key-driven	**Built**
Screen readers	Labels on icon buttons; charts as images with text tables; live region for route changes	**Built** on web, **Partial** on Android charts
Forced colours	Selected capsule outlined in `CanvasText`	**Built**
Text resize	Web reflow to 320 px; Android `sp`; HUD `textScale`	**Built**
Reduced motion	Respect user preference	**Built** (web)
Haptics	Vibration for critical alarms and confirmed actions	**Planned**

13.2 Gloves, noise and outdoor light

- **Gloves.** Field targets are 52 dp or larger on Android task flows and the full width of the screen where possible; the phone-mode tab bar is 92 units tall with 28-unit icons. Avoid swipe-only actions in phone flows: every swipe has a tap equivalent.
- **Noise.** Voice is a convenience, never the only path. Every voice capability has a button or tap equivalent, and answers are shown as text as well as spoken. On Ray-Ban Meta glasses the microphone is the glasses microphone over Bluetooth, which sits closer to the mouth than the phone.
- **Outdoor light.** Dark themes with near-white text (19:1) survive glare better than mid-grey palettes. Minimum sizes outdoors: 16 sp body on Android, 20 units body on the HUD, KPI values at 34 sp or larger. Operators can raise text size through the system setting (Android) or the UI profile (glasses).
- **Safety glasses and polarised lenses.** Do not rely on subtle tints; severity is always paired with a word and, on the HUD, a 2 to 3 unit border.

13.3 Haptics (Planned)

No haptic feedback exists in the code today (no vibration calls on web or Android). The planned pattern: one short pulse when a step is recorded, a double pulse when an action is rejected, and a long pulse for a critical alarm interrupt on Android. Haptics complement, never replace, visible state.

14 Content and voice guidelines

Hydris copy is short, specific and calm. It names the equipment, the value and the next step. It never overstates what the system does.

Table 31. UX writing do and don't

Do	Don't	Why
"Acknowledge" with hint "does not change equipment"	"Fix", "Clear alarm"	Hydris records; it does not operate equipment
"VT-101 vibration high-high · 7.4 mm/s"	"Problem detected on pump"	Name the tag, measurand and value
"Offline · showing data cached 4m ago"	"Error"	Say what still works and how old the data is
"Next step (approved procedure): ..."	"AI recommends ..."	Attribute guidance to the approved SOP
"Send for verification"	"Submit final"	A reviewer and approver still confirm the record
"We couldn't find that record" plus a way back	"404 Not Found"	Plain language with a recovery path
"Ticking checklist items · not permitted by your profile"	Hiding the control silently	Never hide state
"Checking your plant data…"	Silent spinner	Explain waits longer than a moment
"Synthetic"	"Test", "Fake"	Consistent term for simulator data
"Step 3 of 8 · optional"	"Question 3"	Orientation and obligation in one line

Rules:

- Sentence case for every label and title. Uppercase only through the eyebrow style.
- Units follow numbers with a space ("7.4 mm/s"); pH is the exception and leads the value ("pH 6.8").
- Relative times for recent events ("5m ago"), absolute times in tooltips and audit records.
- No em dashes in product copy or documents; use a colon, a comma, a middle dot or two sentences. The guided capture messages currently use a spaced hyphen, which is acceptable on screen.
- Button labels are verbs; one to three words on the HUD.

15 Localisation

The product ships in English only today: the web root sets `lang="en"` (`apps/web/src/app/layout.tsx`) and no string catalogue or i18n library exists on web or Android.

Table 32. Localisation plan

Item	Detail	Status
String extraction	Web: message catalogues per route; Android: `strings.xml` resources	**Planned**
Hindi (hi-IN)	First additional language; Devanagari font fallback for Outfit (Noto Sans Devanagari)	**Planned**
Tamil (ta-IN), Marathi, Gujarati, Telugu, Kannada	Pilot regions in India	**Planned**
Voice	Wake phrase stays "Hey Hydris"; recognition and TTS locale follow the operator	**Planned**
OCR	ML Kit Devanagari model for Hindi labels on audit sheets (per mobile spec)	**Planned**
Numbers and units	Locale digit grouping in tables; SI units unchanged; decimal separator follows locale	**Planned**
Layout tolerance	Labels must survive 40 % expansion; HUD rows truncate with an ellipsis, never wrap into two actions	Rule (applies now)

> **NOTE**
> Indic scripts need taller line heights than Latin. When Hindi or Tamil is added, HUD row padding and the 0.95 line height of `t-kpi` must be rechecked; numerals stay Latin (tabular) in every locale.

16 Dark and light themes

Table 33. Theme support by surface

Surface	Default	Light theme	Mechanism	Status
Console	Dark	Full token set	`data-theme` on `<html>` set by a blocking script before paint; `ThemeToggle` (system, light, dark); stored in `localStorage` key `hydris.theme`	**Built**
Website	Dark	Shared tokens plus `.hx` brand layer	Same	**Built**
Glasses app and simulator viewport	Dark, forced	None (by design)	`FORCED_DARK_RE`	**Built**
Android	Dark	None	`darkColorScheme` only	**Planned** (light scheme for office use)
Word and PDF documents	Light	n/a	Build theme	**Built**

Rules: the blue focal panel, brand blue fills and Alert Orange are identical in both themes; semantic **text** tints are redefined for light so they stay AA (for example critical text `#C41F00` on white, 5.93:1); KPI accent numerals deepen on paper and light surfaces (`kpi-accent-*` overrides).

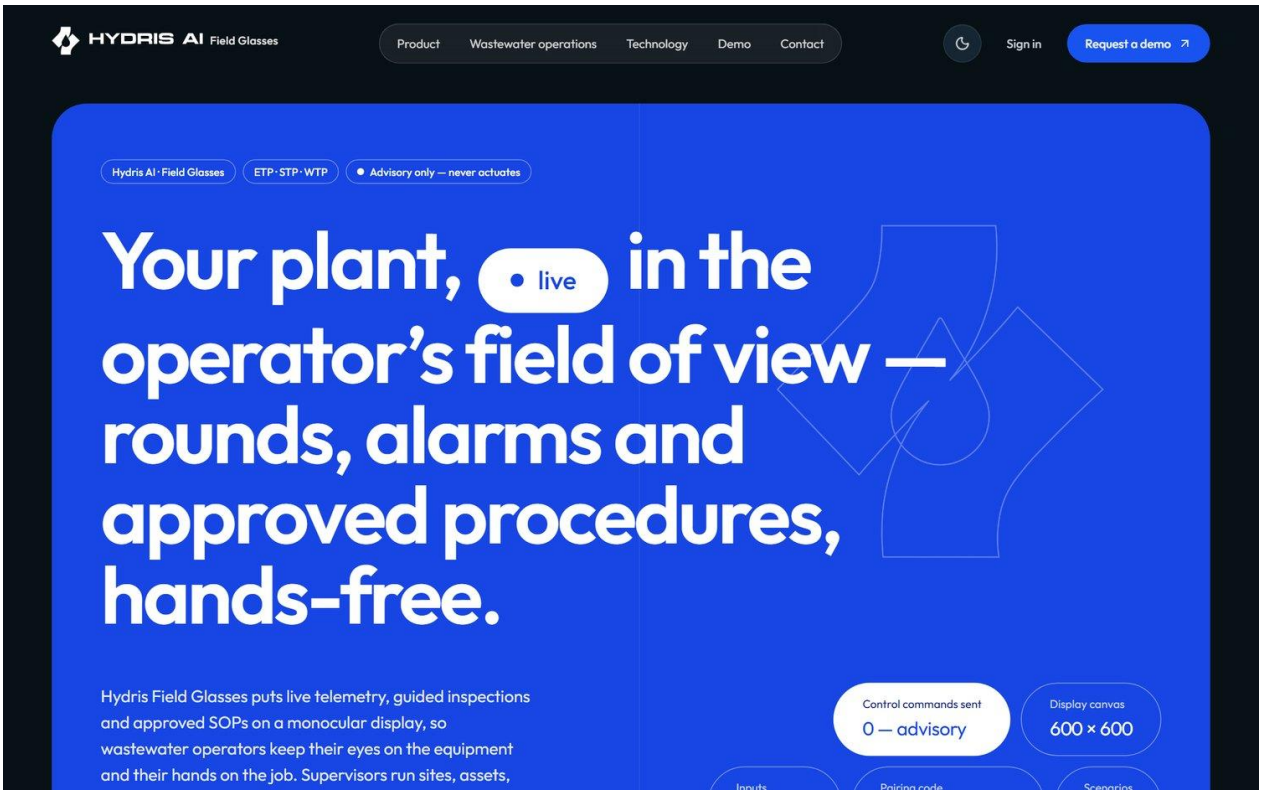


Figure 35. *Website home: brand layer over the shared tokens*

17 Design QA checklist

Use this list on every pull request that changes UI. Items checked here are true of the current code base; open items are known gaps to close.

- ☑ Uses semantic tokens only; no new hex values in components (web)
- ☑ Every value shows unit and freshness, and missing data shows "No data"
- ☑ Severity is shown with a word or icon as well as colour
- ☑ Synthetic data carries the Synthetic marker on every surface
- ☑ Interactive targets are at least 44 by 44 px (web) or 48 dp (Android)
- ☑ Focus is visible and follows the surface colour
- ☑ Charts have an accessible name and an empty state (web)
- ☑ `prefers-reduced-motion` respected (web)
- ☑ HUD screen has one column and one primary action
- ☑ Blocked actions are explained, not hidden (glasses)
- ☐ HUD selected pill, View button and digit cell use white ink on Signal Blue (contrast 5.87:1)
- ☐ Android filled severity chips meet 4.5:1 or use the tinted style
- ☐ Android synthetic chip uses a lighter Insight text tint
- ☐ Android Resolve collects a note before posting
- ☐ High severity colour aligned to `#F26414` across console, website, Android and HUD
- ☐ HUD severity colours moved from inline hex to tokens
- ☐ Android chart canvases have content descriptions
- ☐ Haptic patterns implemented for record, reject and critical alarm
- ☐ Strings extracted for localisation
- ☐ Tested at UI profile text scale 1.3 in both densities and both hands

18 Governance and contribution

The design system is owned by Design, Hydris, with one engineering maintainer per platform (web, Android). Changes follow the same review discipline as code: the Markdown in this folder and the token files are the source of truth.

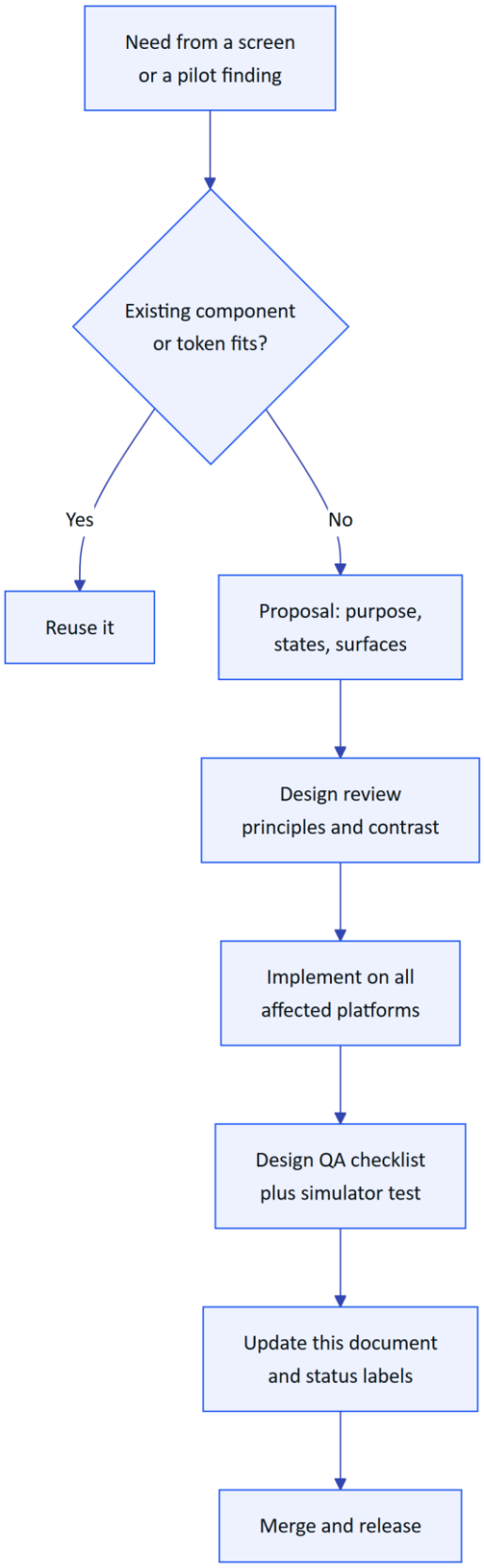


Figure 36. *Contribution process for a design system change*

Table 34. Contribution rules

Rule	Detail
Tokens before components	A new colour, size or radius is added as a token in `globals.css` and `Theme.kt` (and the HUD) in the same pull request
No one-off severity colours	Severity and status colours come only from the severity tokens
All surfaces or a ticket	A change to a shared pattern ships on every surface, or opens a tracked gap with a **Partial** label here
Contrast proof	New colour pairs include their computed ratio in the pull request description
Simulator check	Glasses changes are checked in the simulator at 600 by 600, 640 by 480 and 480 by 480, in phone mode, and at text scale 1.3
Screenshots	Updated screenshots go into `docs/product/images/` (web shots refresh with `node screenshots.js`)
Versioning	This document versions with minor bumps for additions and major bumps for breaking token renames

19 Future work

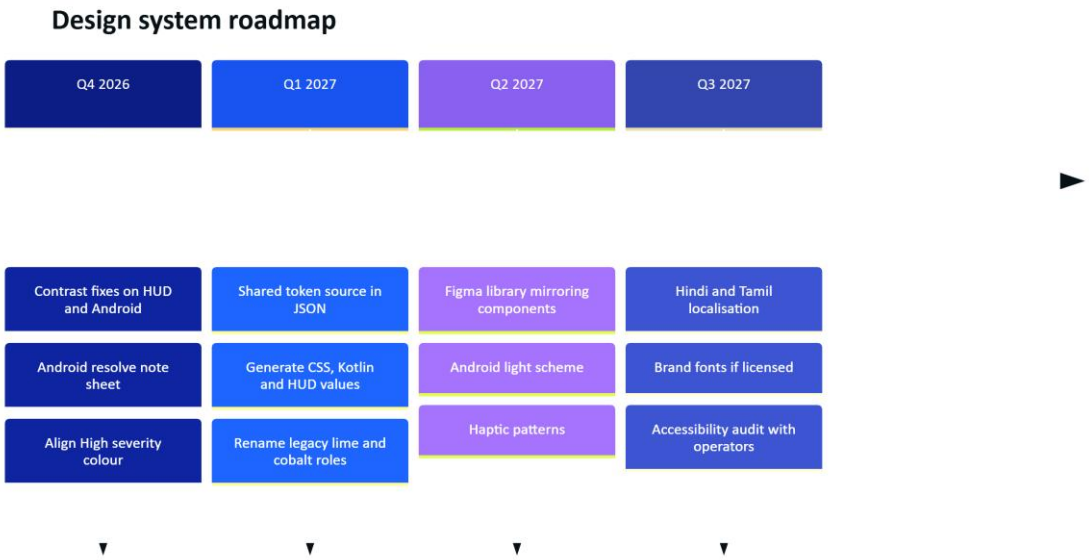


Figure 37. *Design system roadmap (targets, not commitments)*

Table 35. Future work items

Item	Description	Status
Token pipeline	One JSON token source (for example in `packages/domain` or a new `packages/tokens`) that generates `globals.css` variables, `Theme.kt` `Hx` and a HUD palette module	**Planned**
Figma library	Components, tokens as Figma variables, HUD frame templates at 600 by 600 and phone 540-unit width, linked to code	**Planned**
Legacy role renames	`lime` to `signal`, `cobalt` to `purple` across classes and utilities	**Planned**
Android component parity	`StateChip`, `FreshnessBadge`, empty states and a severity chip that matches the web	**Planned**
Push notification design	Critical alarm notification layout and channels on Android	**Planned**
Hardware gesture mapping	Neural band and frame touchpad onto the six-key vocabulary	**Planned**
Field usability study	Glove, glare and noise trials with pilot operators; results to update this document	**Planned**

EXAMPLE

A token entry in the planned pipeline: `color.severity.high = { value: "#F26414", text.dark: "#FF9A5C", text.light: "#A35200" }` generates `--color-high` and `--color-high-text` for web, `Hx.High` and `Hx.HighText` for Android, and the HUD pill and border values, so one change updates every surface.